

Degree project in Mechanical Engineering

Second cycle, 30 credits

Adoption of System Innovations

A Case Study: Feasibility and Viability of Intermediate Storage of Refrigerated Goods Enabling Off-Peak Deliveries in Urban Environments

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Forewords

The project described in this report was carried out by two individuals, Hanna Dahlén and Rasmus Johansson, who are referred to as *the project members* in the text. The collaboration ended at the end of the project, but before the report was finalised. Therefore, only I, Hanna Dahlén, am listed as the author of this report as I am the author of the final version. This means that another report by the second author will be published at some time after this one, which will be very similar as we carried out the same project and parts of the report was written together. Although the project has been carried out by two individuals, future references to any of the reports will refer only to the individual author of each report. Both authors take full credit for their respective reports.

I would like to thank our supervisors, Mats Magnusson and Elisabeth Hörnfeldt, for your support and guidance throughout the project. I would also like to thank Andreij Nylander for your contribution as an unofficial supervisor in the development of our concepts. I would also like to thank Jenny Janhager Stier for your valuable support. Also, I would like to thank everyone else who has provided feedback and guidance along the way and helped us deliver these results.

Stockholm, June 2025

Hanna Dahlén

Abstract

This project has studied adoption of system innovations and problems that can arise in different areas of system innovation. The specific case study has been about goods deliveries with temporary intermediate storage of refrigerated goods to support off-peak deliveries in urban environments. Through iterative concept design processes, four business model proposals and four technical solutions have been developed that can be combined in different ways. The aim of this was to create a modular business offering that can be adapted to customers' specific conditions and needs. Findings have also been produced on the potential for adoption of the concepts by different stakeholders and on their desirability, viability, feasibility, and sustainability, with recommendations for further work. Analyses have been made of the innovation process, the validity and reliability of the project, the contribution of the project results to current research, and the economic, social, and environmental aspects of the project results for society and industry.

Keywords: System innovation, Adoption, Intermediate storage, Transport, Logistics, Off-peak, Goods delivery, Business model

Sammanfattning

Detta projekt har studerat adoption av systeminnovationer och problem som kan uppstå inom olika områden inom systeminnovation. Den specifika fallstudien har handlat om gods leveranser med tillfällig mellanlagring av kylda varor för att stödja leveranser utanför rusningstid i stadsmiljöer. Genom iterativa konceptdesignprocesser har fyra förslag på affärsmodeller och fyra tekniska lösningar tagits fram som kan kombineras på olika sätt. Syftet med detta var att skapa ett modulärt affärserbjudande som kan anpassas efter kundernas specifika förutsättningar och behov. Resultat har även tagits fram gällande möjligheten till adoption av koncepten av olika intressenter och om huruvida de är önskvärda, livskraftiga, genomförbara och hållbara, med rekommendationer för ett fortsatt arbete. Analyser har gjorts av innovationsprocessen, projektets validitet och reliabilitet, vad projektresultaten bidrar med till den nuvarande forskningen, samt vad projektresultaten har för ekonomiska, sociala och miljömässiga aspekter på samhället och industrin.

Nyckelord: Systeminnovation, Adoption, Mellanlagring, Transport, Logistik, Nattleverans, Godsleverans, Affärsmodell

Abbreviations

B2B – Business to Business

B2C – Business to Consumer

HITS – Hållbara och Integrerade Urbana Transportsystem. European project on Sustainable and Integrated Urban Transport Systems (Increasing system efficiency)

IE – Innovation ecosystems

LMD – Last-Mile Delivery

LTV – Life Time Value

NoHa – Norr om Hamngatan

ROI – Return on investment

R&D – Research and development

PSS – Product-Service Systems

TRIZ – Theory of Inventive Problem Solving

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1. Introduction

This chapter presents an overview and explanation of why the project came about, defining the practical and academic motivation, problem formulation.

1.1 Practical and academic motivation

How can the transportation systems of today become more sustainable? This is what a company in the truck manufacturing industry wants to be a part of. Their purpose is to drive the transition to a sustainable transport system. To achieve this, the need for people to make this transition must first be created before it is possible to achieve. In the research project Hållbara och Integrerade Urbana Transportsystem (HITS) that is exploring solutions for the future of sustainable urban freight transport (Closer, n.d.), problems have been identified in certain markets where the potential for this need could arise.

One problem that has been researched in the HITS project is unsustainable traffic flows (Closer, n.d.). Traffic congestion causes several problems to a country's economy and the health of its citizens, such as economic loss in terms of opportunity cost and fuel consumption. Traffic congestion causes problems on an individual level as well, such as time loss, mental stress, and the added pollution that has a negative effect on the environment (Akhtar & Moridpour, 2021). The causes of urban congestion depend on a number of factors including excess traffic, physical capacity limitation, traffic accidents, vehicular breakdowns, roadwork, and poor weather conditions (Chow et al., 2014). Urban goods deliveries are an increasing phenomenon which has become a major contributor to urban traffic congestion, due to delivery trucks having to double-park on the streets when there are no suitable delivery areas (Hammami, 2020). As most urban activities require the movement of goods in some capacity, efficient urban goods transport and logistics is a major factor for urban life and development (Crainic et al., 2023). Verlinde and Macharis (2016) agrees, stating that the growing awareness of the problems caused by urban freight transport requires new innovative solutions. A conclusion drawn in the HITS project (Closer, n.d.) was that one solution to the problem with traffic congestion could be to deliver goods during off-peak hours, which happens during nights and early mornings when there is less traffic. This in turn would ensure that the transition to a more sustainable transport system progresses.

Another problem related to unsustainable transport researched in HITS was that many actors in the retail and restaurants market do not receive their goods on time during the day due to delays caused by traffic congestion in urban areas. This is a serious problem for them as they may not have enough goods in stock for their customers during the day if the delivery is late, especially during holidays or special seasons when there is a higher demand for certain items. However, in HITS it has been identified that many stores and restaurants do not have the right type of storage space to allow for deliveries during off-peak hours when they are not staffed. Storing refrigerated goods is particularly tricky as they must always be kept at certain temperatures (Closer, n.d.). In an attempt to solve this problem, a study of whether off-peak deliveries could be a solution and how it would work must be made.

The HITS project has focused on how to enable off-peak deliveries as it has been stated that it is a choice that is good for the environment and it would be a step in the right direction to

drive the transition to a more sustainable transport system in society (Closer, n.d.). Therefore, the following hypothesis of a proposed solution has been formulated for this master's thesis project: "*Enabling off-peak deliveries in urban logistics through intermediate storage solutions.*" The argument was that if goods would be delivered to small storages conveniently located in urban areas for temporary storage during off-peak hours, traffic congestion in cities would be reduced. The goods from the storage unit would either be retrieved by the customers themselves during the day, or there would be a service that provides the delivery. The question that remained from the hypothesis was: "*Could this be a potential and viable business opportunity for a truck manufacturing company, and what role would they have in the process?*"

An issue that can arise in the context of new innovation initiatives is how to ensure the adoption of all stakeholders, especially in the case of system innovations as they involve a large number of different stakeholders. This project studied some factors that influence the adoption of system innovations by different stakeholders in an ecosystem, as well as the problems that can arise related to these factors and then what managers should take into account to address them. Kristiansen (2024) argues that this is a managerially important issue, as the value of some innovations can be difficult to communicate to customers, and hard for them to perceive before adoption. Therefore, this project aimed to provide new academic research and insights to support companies to better succeed in their system innovation initiatives.

In conclusion, the project has two main purposes, where the first is practical related towards a truck manufacturing company, which will henceforth be referred to as *the reference company* in this report. The second is academically based towards KTH Royal Institute of Technology.

1.2 Problem formulation

This project investigated the feasibility of the hypothesis of the proposed solution to the problems related to urban goods deliveries: "*Enabling off-peak deliveries in urban logistics through intermediate storage solutions.*" Specifically, the project investigated whether temporary storage of both refrigerated and dry goods could reduce congestion caused by heavy goods deliveries in daytime traffic in urban areas. Also, whether it could enable efficient delivery of refrigerated goods during off-peak hours and address both sustainability (social, economic, and environmental) and operational challenges at system level.

Another problem which the project aimed to investigate was adoption toward new system innovations. This research on how to avoid the non-adoption of new system innovations by different stakeholders should be valuable not only for the reference company, but also for other companies in different industries working on system innovation.

1.3 Purpose and objectives

The purpose of this master's thesis was two-fold, consisting of both a practical and an academic research purpose, along with practical and academic research objectives. The practical purpose was to investigate the potential for novel business offerings for the reference company. The intended objectives were to offer validated concepts for business models and technical solutions which enable temporary intermediate storage of goods for off-peak deliveries in urban environments. In addition, valuable conclusions on the feasibility, viability, desirability, and sustainability of the concepts, and in turn identify whether this could be a valuable business that the reference company should pursue.

The academic research purpose was to contribute to the existing knowledge in the theoretical areas of system innovation, concept design, and customer segmentation. The academic research objectives were to provide novel research on adoption toward new system innovations, and to investigate how concept design and customer segmentation relate to system innovation.

1.4 Research questions

From the project description and purpose, the study aimed to answer the following three research questions:

- RQ1: *How desirable, viable, feasible, and sustainable is temporary storage of goods for the reference company to develop and implement?*
- RQ2: *What are the critical factors and stakeholders that affect the proposed solution?*
- RQ3: *How can adoption among stakeholders of system innovations be facilitated?*

1.5 Delimitations

Several delimitations were made during this project mainly due to the amount of time and resources. The master's thesis project was limited to a time frame of 30 credits per individual, which means approximately 80 hours per week in a 20-week period. Another delimitation was that the developed concepts were designed according to the geographical area of Stockholm. Research was conducted on other areas and actors from other cities were also interviewed, but the applicability of the concepts was delimited to Stockholm. Another delimitation made was not to focus on road transport and logistics as a whole, but only a more narrow focus on goods deliveries.

Delimitations of the scope were made as the project progressed. Initially the scope was very large and could be taken in many directions, but due to the iterative process and factors affecting progress, the project was narrowed down and some activities were not carried out as initially planned. These activities included accurate ROI calculations for the reference company and their customers, as well as visual customer journeys for each customer segment. More interviews with more municipalities, more logistics companies, and other relevant individuals with knowledge of logistics and system innovation were also an ambition, but these were not prioritised and therefore not carried out.

Another delimitation was made within the literature review of system innovation, specifically for adoption, where internal adoption within an organisation was not considered. This was because the innovation in this project involved more than just one organisation, but more parts of society, meaning that internal adoption was not as relevant to analyse.

2. Theoretical framework

This chapter presents the current state of knowledge from literary sources on relevant theories of system innovation, concept design, and customer segmentation, all of which are topics relevant to the project. System innovation was the type of innovation worked on in this project since the project involved many different actors, and major changes were created at the system level. The topic focus was on adoption of system innovations and what might hinder it. Therefore, a literature review was conducted to provide more background information on the topic. The project involved a lot of conceptual work and therefore a literature study was carried out on this topic and how it relates to systems innovation. The conceptual work included, as an important part of system innovation, customer segmentation, which also contributed to the inclusion of the topic in the literature review.

2.1 Background to system innovation

This section addresses the topic of system innovation with a summary and definition of the concept, as well as an explanation on the adoption of system innovation and how it differs from other types of innovation. It also provides an in-depth review of the different areas of system innovation and the problems that can arise within them that can hinder adoption.

2.1.1 Definition of system innovation

System innovation is a process in which multiple actors, such as firms, customers, authorities, research institutes, etc. collaborate and depend on each other when contributing to the development of new innovations. Interconnected institutions create, store, and transfer knowledge and skills to create new technologies (van Mierlo et al., 2010). There are three main type of actors that usually collaborate in this type of innovation systems: industries that aims to generate economic wealth, universities/research centres that aims to develop new knowledge, and governments that want to control and keep their society attractive for all actors (Leydesdorff & Ivanova, 2016; Verlinde & Macharis, 2016). The relationship between these three actors is often referred to as the *Triple Helix model*, where interactive cooperation between all three actors leads to the achievement of their long-term strategic goals, according to Razak and White (2015).

System innovations include product and process innovations, but also changes in user practices, markets, policies, regulations, infrastructures, lifestyles, and business management. These system innovations significantly impact the innovation efforts aimed at achieving societal sustainability (Gaziulusoy & Brezet, 2015; Geels, 2005). System innovations also differs from normal product and process innovation regarding scale, speed, and involved institutions (Adamides, 2023). A system innovation often consists of different *socio-technical systems*, which are a set of societal functions and elements that are interconnected in different ways, such as technologies, regulations, infrastructure, user practices, etc. The term highlights that technological and social aspects are strongly interconnected, and a system innovation often involves a transition between different socio-technical systems (Geels, 2005).

In addition, system innovations are often linked to *wicked problems* (Kuokkanen et al., 2016), which are complex social issues in the social system that are ill-defined (de Tarragon et al., 2024). These system innovations usually lead to major changes in the whole production and

consumption systems, which in turn affect the material flows and their institutional structures, but also the behaviours of the different actors involved (Kuokkanen et al. (2016).

2.1.2 Adoption of system innovations

The adoption process of innovations, as defined by Anand et al., (2016), "*refers to the sequence of stages through which a consumer progresses from their first awareness to the final acceptance of the innovation.*" The authors argue that adoption of new innovations is very difficult to achieve as perception of individual buyers play a crucial role in the innovation diffusion process, which is the process of communicating an innovation through different channels. Fadwa et al. (2023) argue that successful implementation of urban innovations specifically requires addressing and overcoming opposition from involved stakeholders. In turn, it could increase the stakeholders' willingness to adopt urban innovations. Therefore, it is crucial to reduce resistance by implementing structured processes for innovations.

Raising awareness of a product in the market is very important because it provides individuals with information about the innovation's characteristics. An individual can more easily adjust their thoughts about an innovation if there is a lot of available information about it that the market can access. The attitude of individuals towards an innovation is one of the main factors leading to either adoption or rejection. In a well-informed market about a product, there are usually individuals who are likely to be motivated by the product and adopt the product early on and may in turn influence others to purchase the product as well (Anand et al., 2016).

Adoption usually starts with the realisation that there is a need for something and then moves on to finding solutions. Next, the first decision is made to try to adopt a solution and lastly, to the final decision to try to proceed with implementing the solution (Wisdom et al., 2014). The authors further argue that the adoption process on a system level is complex. One particular challenge is to promote change in routine practices that decision-makers do not perceive as necessary.

2.1.3 Different dimensions of system innovation and related problems that might hinder adoption

System innovation is a very broad topic that exists in several areas and is affected by many aspects. In this project, literature was compiled into five dimensions that were identified to include aspects commonly addressed in system innovation and that were particularly related to the project. The dimensions were: *ecosystems*, *acceptance and legitimacy*, *governance and regulations*, *business models*, and *technology*. It was identified that these dimensions were critical factors in the introduction of new innovations and that they have a high impact in adoption by all involved stakeholders, who can be affected in different ways by the shifts that occur when new innovations are introduced. This section presents a literature review of these dimensions and how they relate to system innovation, as well as the correlation to innovation adoption. The focus was not on adoption within organisations, but rather on system innovations involving multiple actors in society and how adoption could be achieved by them.

There are several challenges and issues that can arise within these five dimensions, which organisations and governments must address to succeed in their innovation efforts. Therefore, this section also includes a review of the literature on these issues and how they relate to what may hinder the adoption of new system innovations when multiple stakeholders are involved.

Ecosystem

The concept of innovation ecosystems (IE) is based on viewing innovation as a complex phenomenon where different actors with distinct characteristics and competences need to interact and collaborate with each other (Hoffmann et al., 2022), and together they create a value proposition for the ecosystem (Thomas & Ritala, 2021). Typically, they are orchestrated by a central actor that possesses key technology or other resources (Thomas & Ritala, 2021). Implementing the ecosystem perspective provide a systemic approach on innovation by focusing on how a network of actors creates and sustains competitive advantage. Moreover, the actors in the system are not hierarchically governed but rather act towards their own goals (Suominen et al., 2019). However, the Tripel Helix, i.e. industries, universities, and governments, often take a more leading role in innovation ecosystems (Leydesdorff & Ivanova, 2016).

To create value propositions, innovative companies must often rely on other actors in their innovation ecosystem, which brings several challenges for the company's managers (Talmar et al., 2020). Hoffmann et al. (2022) point out that since there are multiple actors with different interests and professions in ecosystems, the challenge arises to co-create a value proposition within a collaborative network. Managing such a network is also challenging due to the variety of actors involved, power and information imbalances, conflicting interests, and different perspectives on how to achieve common goals. Additionally, the authors argue that interactions become complex but also crucial as they combine both cooperation and competition. Kuokkanen et al. (2016) argue that a shared vision among the actors involved is important in system transitions. This is because systemic change needs the support of key actors and has the potential to provide a basis for overcoming problems when there are no established rules or institutions.

To address these issues, academic research has resulted in a list of factors that managers can consider. For example, they should consider the modularity of their ecosystem and the respective value creation structure, and in order to manage the large number of actors involved, the specific roles of actors within the value structure should also be considered (Talmar et al., 2020). It is also important to notice that studying a system or market in isolation can be biased, as a single approach may be inadequate when explaining the system/market dynamics. Therefore, there is a need for a cross-sectoral approach (Anand et al., 2016).

Acceptance and Legitimacy

A definition of acceptance formulated by Fadwa et. (2023) is "*It is the fact that a person declaring that he subscribes to the undertaking offer which is proposed to him: it constitutes the apparent mark of consent. Acceptance can be express or implied.*" Alsheimer et al. (2025) express that existing research on acceptance of innovations has largely had a focus on individuals' preferences. Dunphy and Herbig (1995) argue that of all the factors required for successful introduction of an innovation, understanding customer needs are the most important one. Companies need to understand why customers accept an innovation or not in order to be more effective in their innovation efforts and find ways to improve competitiveness and profitability.

For an innovation to be accepted in small contexts, Johnson et al., (2006) argues that the actors must link it to the existing, widely accepted cultural framework of beliefs, values, and norms.

As a result of the innovation being successfully justified or accepted by a small group of stakeholders, it receives validation. This in turn leads to the creation of a new prototype as an acceptable way of doing things to fulfil local needs or goals. Furthermore, the authors argue that when the innovation can be scaled and diffused, it often does not need to be justified as clearly in new situations as it did initially in the smaller context where it was introduced and could achieve adoption.

A definition of legitimacy created by Schuman (1995) states that "*legitimacy is a generalized perception or assumption that the actions of an entity are desirable, proper, or appropriate within some socially constructed system of norms values, beliefs, and definitions*". Organisations seek legitimacy and should therefore consider the individuals who influence and are influenced by the organisation's activities (Agewall & Sommarström, 2014). If legitimacy emerges, it may mean that society has a better understanding of the organisation's activities and its reasons for existence.

Thomas and Ritala (2021) argue that when the organisational structure of ecosystem emerges, legitimacy is required to overcome the *liability of newness*. Additionally, the authors argue for the importance of the particular roles that different actors have in an emerging ecosystem in terms of legitimacy. For instance, users and other external actors such as regulators, competitors, and the media have a major role in facilitating legitimacy in an emerging ecosystem. According to Bunduchi et al. (2022) there are different ways of acquiring legitimacy among stakeholders. The most common strategy involves conforming to stakeholder norms and expectations. Other strategies include selecting a group of stakeholders that has a similar attitude and is more likely to provide legitimacy or influencing the environment by acting to change a stakeholder's attitude and creating a new type of supportive environment with new rules, practices, and norms.

Achieving legitimacy in ecosystems is challenging because ecosystems are made up of complex systemic interdependencies between different actors that are not linked by employment relationships. And since achieving legitimacy is particularly important when new ecosystems are created, a key question to address is how an emerging ecosystem can be perceived as legitimate by the participants in the ecosystem, as well as other actors in the wider environment (Thomas & Ritala, 2021). One social aspect that is important to consider is assumptions about the consequences and possible future outcomes of different activities. According to Berkhout (2006), this is particularly important in the case of innovations that intend to influence and change the world, as is often the case with large-scale system innovations. Expectations are crucial to all social activities, and it is widely accepted that they influence the attitudes and behaviours of social actors. Making assumptions about these expectations is therefore crucial to being innovative.

Both acceptance and legitimacy play a major role in technology diffusion, as they influence the success and speed of the diffusion and siting processes (Alsheimer et al., 2025).

External governance and regulation

Local authorities face the challenge of making urban goods logistics more sustainable, due to current guidelines determined by sustainability objectives. Therefore, it is necessary to involve various stakeholders of the logistics ecosystem to facilitate sustainability efforts (Przybylska

et al., 2023). Furthermore, the authors argue that the successful implementation of new solutions in urban freight transport depends to a large extent on the openness and acceptance of stakeholders. It is important to know and understand the interests of the different stakeholders in order to successfully achieve sustainability.

Wisdom et al. (2014) argues that in the pre-adoption stage, policy, legislation, and regulation have a positive relationship with innovation adoption. Later, during the adoption phase, further legislation, regulators, policies, culture, and accreditation standards are also associated with increased use of innovations. Johnson et al. (2006) argue that this type of surrounding environment also has an impact on legitimacy, which is an important part of adoption. Regulatory legitimacy comes from actors who have authority over organisations, such as government regulators. These define what is legally or procedurally acceptable through requirements and sanctions (Johnson et al., 2006).

In the context of market failures, Kuokkanen et al. (2016) argue that governments can be necessary to correct inefficiencies. This is particularly the case when it comes to environmental issues, where external effects and the over-utilisation of resources occur. The authors argue that governments can assist in correct market failures by implementing factors such as taxes, subsidies, and tradable permits. However, in the case of system innovation, market failures alone are not sufficient to justify this type of policy action.

Business Models

Business model innovations are key to creating sustainable businesses, but companies have been reluctant to adopt them. Organisations depend on system-level adoption in society and can only be sustainable if the whole system is sustainable (Laukkanen & Patala, 2014). System innovation can be created in the form of a transition from an unsustainable system to a sustainable one. In these cases, it is often considered an effective method for creating solutions to the challenges facing society. This is because there is a strong focus on the complex and multidimensional nature of the issues (Kuokkanen et al., 2016). Over the last decade, knowledge on system innovation and transition for sustainability has increased, with a strong focus on business models and less on design and product/service innovation in the context of system innovation and transition for sustainability (Gaziulusoy, 2015).

Technology

System innovation consists of many different elements and actors, but the name of the system is often based on the main technologies involved. This is often due to the tendency to focus first and foremost on concrete objects, but also to the fact that technology often plays a very important role in system innovation (Geels, 2005). According to Faissal Bassis and Armellini (2018), technology development takes place in a complex interaction between different stakeholders, such as industrial actors, investors, entrepreneurs, researchers, and governments. Geels (2005) further argue that technological innovation is often what companies compete for. Technological innovations can also affect socio-technical systems in different ways and be a crucial part of the transition between them. This means that technology is likely to play an important role in many system innovations.

A problem in the design of a socio-technical system is to understand the requirements of the software component and how people and organisations are affected by introducing technology.

In a socio-technical system, the human, organisational, and software actors are highly interdependent in order to achieve their objectives. Therefore, an important part of the design of a socio-technical system is to identify how the actors are interdependent, and whether the requirements can be met by all within the socio-technical system (Bryl et al., 2009).

Historically, software and information technology organisations have spent little resources on technology adoption, and rather focused on aspects related to the deployment of the technology (Prasad & Nori, 2008). Flynn (2007) highlights the complexity of assessing public acceptance of new technologies. They argue that it can be difficult for citizens to express their views and give their approval before the technology is introduced. Companies often conduct extensive market research when developing new technologies, but sometimes this can be insufficient to assess public attitudes towards the technology. Although customers' willingness to pay is often considered an indicator of acceptance, such decisions often involve trade-offs, which can make it difficult to verify real attitudes or preferences. Bassis and Armellini (2018) adds by stating that regulations and legal compliance are some complexities that the global market adds to the emergence of new technologies.

Market failures are another large problem related to new technology as they can prevent optimal resource allocation due to factors such as invisibilities, lack of usability, and uncertainties. Market failures in technology and innovation can lead to underinvestment in basic research and insufficient funding for research and development (R&D), which is problematic as these are essential factors to knowledge creation and innovation. (Kuokkanen et al., 2016).

2.2 Concept design in system innovation

This section presents the process of designing a concept in system innovations. A definition of concept design is presented, following the general phases of concept design, and the design cycle. Finally, commonly occurring frameworks and concept design specifically in system innovation are presented.

2.2.1 Definition of concept design

The definition of concept design is according to Taura & Nagai (2013b) "*the process of composing a desirable concept towards the future*". This is an abstract definition in both the objective sense (what is desirable?) and in the temporal sense (how does the future look like?) with the benefit being a high degree of freedom.

2.2.2 Phases of concept design

Concept design is the process of composing a desirable concept towards the future, as the definition suggests. The process consists of a few phases which are meant to be cyclical and iterative to be effective. These phases are called: *Explore*, *Create*, *Evaluate*, and *Manage*.

Explore: What are the needs?

The first phase of generating a concept is the explorative phase which involves gathering data about market needs. The main purpose of this phase is to understand the criteria that the innovation has to fulfil (University of Cambridge, 2024). Need identification can be done by doing a thorough desktop research, interviews, and observations in order to map the existing products, services or Product-Service Systems (PSS) and produce a basis on what the concepts

can build on in order to differentiate from the existing market options enough to be a desirable, viable, feasible, and sustainable novel offering. Different types of stakeholders might have different needs and requirements, so when the needs for end users are identified, it is important to look at the needs of all of the involved stakeholders.

Create: How can the needs be met?

When data has been collected, the create phase should be initiated. This phase is quite extensive and consists of ideation grounded in the research findings, developing complete solutions, and making demonstrative prototypes (University of Cambridge, 2024). This is both an explorative and convergent effort. The function of ideation is to stimulate ideas and produce as many ideas as possible which might solve a need. After ideating, the next step is formulating a concept, by picking out one or many ideas and formulating a complete solution. This phase is a convergent phase where the complete set of ideas are either approved and used or discarded. The key function of this step is to identify the best solutions to the identified problems.

After formulation of the concepts, prototyping effort are the next step which involves producing inexpensive versions of the concepts to be able to test them. This is preparatory work for the next phase of testing and evaluating the concepts. In the create phase, it is encouraged to strive for simplicity. The main function of this is to reduce unnecessary functions in the concepts. Assumptions should be challenged to not get stuck in how problems are solved today. The last important aspect to consider in this phase is to give all ideas a chance regardless of how unfeasible they might feel.

Evaluate: How well are the needs met?

The evaluation phase is used to evaluate the validity of the concept for all of the involved stakeholders. It is an intensive test phase where the function is to generate feedback on the formulated concept and the validity of the assumptions made from the research, in order to learn and improve. To make the concept design cycle, the next step is to return to the explore or create phase, depending on the feedback and the adjustments which have to be made. This phase is especially important early on in a project as early frequent feedback is vital when there is still room for meaningful adjustments. It is also stressed that creativity and a critical view is good in order to find meaningful issues with the concepts (University of Cambridge, 2024).

Manage: What should we do next?

The manage phase is not sequentially next to evaluate but rather a stage which should always be operating in the background. The main purpose of the manage phase is to review the progress and to plan the next steps, keeping the process on track and on budget. Another important management focus is to ensure good communication with internal and external project partners. It is also highlighted that planning to be flexible is crucial to allow for unaccounted game-changing ideas along the process (University of Cambridge, 2024).

2.2.3 Frameworks in concept design

Within concept design and the general phases of explore, create, evaluate and manage, there are a few frameworks and tools mentioned in the literature to be useful and they have their own set of useful characteristics. There are also some specific frameworks mentioned in the literature of concept design which are especially useful in system innovation contexts.

Design Thinking is a framework which fosters creativity, innovative thinking, and has an emphasis on collaboration (Gheorghe et al., 2021). The phases which this framework entails are especially useful for effective problem solving in product development, making it commonly mentioned in concept design contexts.

Another framework which has been especially specialised toward collaborative creative problem solving in systems is called the *planetary model* (Steiner, 2009). Shortly it is a categorisation of components within concept generation and how they interact and relate to one another in a larger system. In system innovation the goal is to address complex challenges and the planetary model aids in visualising the dynamic nature of innovation and the relationships of the concept generating components like *Stakeholder management*, *Problem finding*, *Objective finding*, and *Generation of Alternatives*.

Double Diamond is a structured approach framework complementary to Design thinking which facilitates discovering opportunities through divergent and convergent thinking (Marin-Garcia et al., 2020). In concept design, this encourages exploration of multiple solutions before narrowing down to the most viable ideas, fostering creativity and innovation. The versatility of the framework makes it a valuable tool for addressing design challenges in different contexts.

TRIZ (Theory of Inventive Problem Solving) is a framework particularly valuable in innovation and problem-solving. It involves a systematic approach to problem-solving with the emphasis on identification of contradictions within a problem (Cerit et al., 2014). In a concept design context, this method allows moving from problems to final result by systematically analysing contradictions within problems to be solved which may hinder progress.

2.2.4 Concept design in system innovation

Concept generation is important to enable creativity at an early stage of design. Design in societal contexts is crucial to successfully innovate systems, to generate higher efficiency but also well-being (Taura & Nagai, 2013b). In system innovation, concept design is a lot more challenging due to collaboration necessities. There cannot be a focus on only smaller parts of the system, but rather a holistic view is necessary (Steiner, 2009).

It is important to regard collaboration, strategic planning, and a balance between feasibility and creativity to develop successful innovative concepts in a complex environment. Other companies in the value chain as well as external potential partnerships, even competitors, have to be involved in a cooperative manner in order to develop innovation of entire ecosystems (Steiner, 2009). Firms therefore play a crucial role in navigating innovation spaces and generating novel concepts as they create opportunities in technology, markets, and productiveness, with internal competences such as technological knowledge, customer knowledge, and business knowledge (McKelvey, 2016).

It is important to implement design cycles over time to ensure that a product or service offering is maintaining its relevance in the market. Through iterations, a better understanding is built of the needs, overall better solutions, and stronger evidence through more validation and data (University of Cambridge, 2024).

PSS is closely linked to system innovation and is common today as a way to design a business model to satisfy customer needs regarding the utility of a product, and its connected services.

Traditional business models on the other hand are focusing on that customers primarily demand products specifically (Sojung & Byungun, 2012). The concept design in these types of systems involves both technical product concepts but also business model concepts which together create the overall PSS concept. The complexity lies in the number of different ways to offer a product and service in a system of actors, to minimise adoptability difficulty.

Concept generation can be separated into First-Order concept generation, related to problem-driven phase, and High-Order concept generation, related to inner-sense driven phase (Taura & Nagai, 2013a). The key differences between these are the complexity and the abstraction level of the concept design. In system innovation, both orders of concept generation are important, as it is necessary to address both the practical functional aspects as well as the broader conceptual aspects of a system. This ensures that the "*what*," "*how*," and "*why*" of the identified problems are answered and that future challenges are anticipated.

2.3 Customer segmentation

This section presents the topic of customer segmentation, which can be an important part of all types of innovation to identify customer value. Here it is specified why it is important to conduct a customer segmentation and how to do it.

2.3.1 The importance of customer segmentation

Customer segmentation is defined as when a market is divided into different groups of customers with various needs, characteristics, or behaviours, and who may require different products or who may have different responses to marketing efforts. Customer segmentation has great potential as a tool as it can enable companies to optimise their marketing and development of new products (Cooil et al., 2008). Customer segmentation is most used in the e-commerce industry, as retailers are constantly looking for ways to expand their market, while maintaining a stable relationship with their regular customers (Gupta et al., 2023).

A company's success depends on its ability to build and maintain customer relationships, which is why it is important to build strategies for customers based on their different values (Kim et al., 2006). According to Hosseini and Shabani (2015), customer relationship management can be seen as the key to achieving revenue growth and profitability, as today there are rapid changes in marketing that shift the focus from the product to the customer. Alves Gomes and Meisen (2023) agrees by saying that it is essential to develop an understanding of an individual customer's interests, needs, and motivations. Furthermore, Alves Gomes and Meisen confirm that customer segmentation is a frequently used method for this purpose.

2.3.2 How to conduct a customer segmentation

There are several methods on how to conduct customer segmentation. Demographic segmentation has been the most traditional method, but newer approaches have been developed that also considers customer's attitudes, motivations, and preferences (Marcus, 1998). Gupta et a., (2023) agrees with Marcus that the most traditional approach is demographic customer segmentation, but they have in their research discussed psychographic, behavioral, and geographic segmentation, which focuses more on subjective attributes of individuals. A more simplified way of separating customers has been discussed by Ozan (2018), who states that the most common way to conduct customer segmentation is to separate one customer from another by calling a group of customers as premium and the rest as standard.

There are other types of methods that are related to more specific aspects of the individual buyers. Cooil et al., (2008) suggests that customer segmentation can be based on five different types of segmentations systems: value, behavioural, lifestyle, lifecycle, or activity. They can be used individually, but usually several are combined. Other methods based on value are discussed by Kim et al. (2006) who proposes three methods where Life Time Value (LTV) is the key type of value to take into consideration, where LTV has been defined by researchers as the total revenue generated from a customer, minus customer-related costs. Marcus (1998) proposes another approach, the Customer Value Matrix, which is especially useful for small retail and service businesses.

Customer segmentation can also be done using AI tools and algorithms. Alves Gomes and Meisen (2023) has identified in their research that the most commonly used segmentation method is k-means, as it is an established approach that is easy to implement. The goal of k-means is to group gathered data points into segments with minimised distance between the data points.

3. Research design and methodology

This chapter presents the structure of how the study aimed to conduct research to answer the research questions, and the methods and processes that were used to do so.

3.1 Overall research approach

An iterative innovation process was used during the practical part of the project. There was continuous planning and evaluation of the progress, during which alterations were made to maintain continued progress. The *Design thinking* process model was used, as described in more detail in Section 3.2. Since the project consisted of many interdependent parts that influenced one another, the Design thinking model was modified and extended with three sub-phases to better suit the nature of the project. Within these sub-phases, various activities were conducted, not in a strict sequence, as some needed to be conducted in parallel. Therefore, the activities within the phases were not presented in a strictly chronological order in this report.

For the academic part of the project, the literature presented in the theoretical framework was complemented and received contributions from the practical parts of the project.

3.2 Process models and software programmes

This section provides an in-depth description of the innovation process used to conduct the research and to perform all the activities involved in the project. In addition, the different software programmes used as tools during the project and what the purpose of their use was, is described in this section.

3.2.1 Modified Design thinking

Design thinking is an iterative non-linear process with a customer-centric approach for identifying and solving problems (Luchs, 2015). It is an analytical and creative process that is useful for experimenting, creating prototypes, gathering feedback, and redesigning (Razzouk & Shute, 2012). This process model was initially chosen for this project because it was presumed that the customer-cantered approach was likely to be useful when examining the benefits that the proposed solution would bring to all involved stakeholders. However, it was realised quickly that the standardised framework did not suit the nature of the project very well. Therefore, the model was modified to better suit the project and personal preferences of how to conduct research. Three new sub-phases were created: *research phase*, *innovation phase*, and *validation phase*, and these sub-phases were incorporated into Design thinking. Within the new phases, several activities were carried out in parallel. The division of how the sub-phases was incorporated into the Design thinking phases, as well as the activities specific to each sub-phase is visualised in Figure 1. The activities are described in Section 3.3 – 3.5.

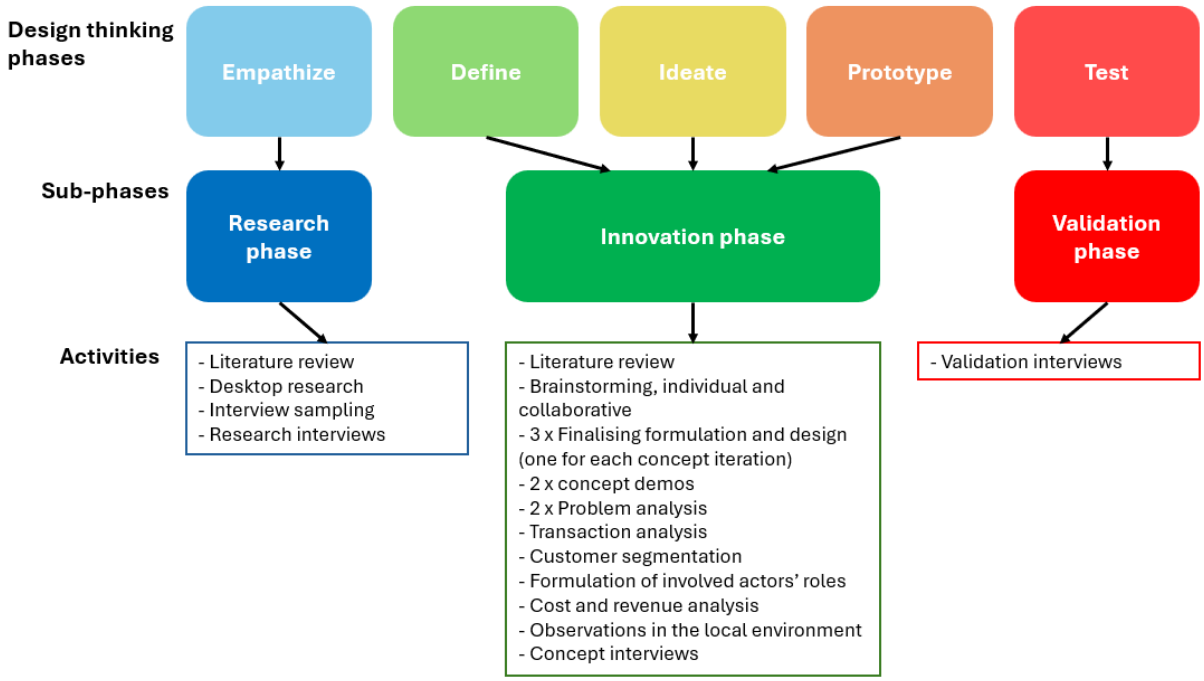


Figure 1. Design thinking phases, developed subphases, and activities. Note. This figure is an altered version of the original produced by Stanford d.school. Copyrights 2012 by Stanford d.school.

Research phase

The goal of the research phase was to gather knowledge and information from several perspectives in order to lay a foundation for later stages of the project. The activities conducted in this phase was: a literature review, desktop research, interview sampling, and semi-structured research interviews with primarily open-ended questions. An explanation of the activities can be read in Section 3.3.

Innovation phase

The innovation phase was the largest phase of the project, during which the conceptual work took place. Three concept iterations were developed, and various activities were conducted during these processes. Observations in the local environment were also conducted to identify needs, problems, and opportunities related to the developed concepts. During this phase, semi-structured concept interviews were conducted, but this time with more leading questions in order to gain insights from potential customers and other stakeholders who contributed to the concept development process. An explanation of the activities can be read in Section 3.3 and Section 3.5.

Validation phase

The final phase, the validation phase, aimed to provide insights into whether the finalised concepts had any potential for being realised and whether there was interest among the respondents to potentially invest in them if they were to be realised in the future. Therefore, several validation interviews with primarily leading questions were conducted. More on these interviews can be read in Section 3.3.

3.2.2 Modified Sprint

A *Sprint* is a fixed period of time, usually between one week up to a month, where specific work has to be planned, executed, and made ready for review. When the time period has ended, the Sprint ends and the next one can start, which follows the same structure. A Sprint optimally consists of five stages: *Define challenge and target*, *Sketch detailed solutions*, *Decide on the best solutions*, *Prototype*, and *Evaluate and learn* (Knapp, 2016).

Sprint was used in this project with the purpose of facilitating an iterative process where the progress could continuously be evaluated. The process described by Knapp (2016) was used partially; however, the stage headings and the purpose of each stage were adjusted to better fit the project. The headings were *Sprint planning*, *Execution*, *Sprint review*, and *Sprint retrospective.* Otherwise, the traditional framework was used throughout the project with a time-period of two-weeks, where the project was planned, executed, reviewed, and looked at in retrospect. But, due to the fast pace of the project and since it was very sensitive to changes, which occurred on a continuous basis, it was determined that two weeks were a too long period of time for planning, and there was a need for more short-term plans with more concrete tasks. Therefore, a jointly and individual weekly to-do lists were added that were updated continuously. The semi-weekly meetings were still held, where the progress was assessed as a complement for the new structure.

3.2.4 Software programmes

A couple of software programmes were used during the project for various purposes. All programmes were used ethically, and only to assist with the project work that would be difficult to conduct manually.

Microsoft Teams was used for several reasons. First, the plan for the project and the various tasks were formulated in the tasks section of the software. The project members were easily able to check mark which tasks were completed and formulate new ones, which facilitated a clear management structure. The second reason was to have digital interviews and meetings were the recording and transcription feature were used to easily look back at what information was provided.

The second programme that was used was *ChatGPT* by OpenAI. The purpose was two-fold. First, it was used to help correcting the grammar in this report. Second, it was used to create AI-visualisations of the concepts to provide realistic realisations to help with clarity during presentations and in this report.

Another programme was used for writing purposes was *DeepL*, which was also used to assist with correcting the grammar in this report.

The last programme was *Mural*, which was used to create a collaborative workspace used during a workshop which was held during the innovation phase. How Mural was used is further explained in Section 3.5.2.

3.3 Data collection

This section presents how the research was conducted and what methods were used to collect various types of data.

3.3.1 Theoretical literature review

During the research phase, a comprehensive literature study was conducted with a systematic review of academic journals, industry reports, and white papers to build a theoretical foundation for the project. The theoretical literature review was done by searching existing academic sources on *system innovation*, *innovation adoption*, *customer segmentation*, and *concept design*. Emphasis was placed on using recent and relevant frameworks and models related to the research topics. Search engines, such as *KTH Primo* and *Google Scholar*, were used to search for articles using keywords such as *system innovation*, *adoption*, *acceptance*, *legitimacy*, *ecosystem*, *customer segmentation*, and *concept generation*.

3.3.2 Desktop research

The desktop research involved a literature review of academic and research journals, well as searches for information on company websites, which comprised a market analysis. The purpose for conducting a market analysis was to identify the key issues related to urban logistics, and the needs of all stakeholders affected by the proposed solution. It also facilitated the assessment of potential demand and the suitability of the target market. A market analysis was conducted on existing solutions similar to intermediate storage and urban transport systems in urban environments. Therefore, the areas of *urban logistics*, *goods deliveries*, and *storage of goods* was researched.

3.3.3 Interview sampling

To support the research on system innovation and to gain insights into urban goods deliveries relevant to the project, several interviews were conducted with individuals from diverse backgrounds. A strategic sampling method was used to select suitable interview participants, based on recommendations from both the academic and industrial supervisors of the project, as well as on the project members' own research into relevant companies and organisations.

The aim of the interviews was to obtain a wide range of insights and perspectives from individuals with different skills, professions, and expertise, who could contribute to several aspects and phases of the project. This approach was motivated by the need to understand the complexity and large number of stakeholders typically involved in system innovation ecosystems in general, and more specifically in the field of heavy urban transport.

3.3.4 Interviews

Several interviews were made for qualitative research. They were mostly semi-structured with a written interview guide tailored to each individual interview. The Teams software was used in order to record and transcribe the interviews, which was later used in order to analyse the interview data. Three different types of interviews were conducted: *research*, *concept*, and *validation*.

Research interviews

The research interviews were conducted in the research phase in order to gather insights related to the theoretical research and for the concept development. These interviews were semi-structured with mainly open-ended questions where the main goal was to gather insight in the contextual setting of the project: logistics, how the logistics supply chain works today, how partnerships are set today, and what actors are involved and how in the entire system. There was also the academic research goal to gain insights from different industries on their

perception and experiences of system innovation and innovation adoption. Some preparations were made by looking at annual reports of the companies to interview to try and deduce how money is spent and to formulate relevant follow-up questions to assumptions.

Concept interviews

The customer insight concept interviews were early concept validation interviews regarding the first iteration of conceptual solutions. The function of these types of interviews were to triangulate the accuracy and the validity of the concepts and reducing the scope moving forward in order to continue with the concept design and creating fully formulated business models to match the concepts, in the most accurate way possible.

Validation interviews

The final validation interviews were made by presenting the finalised concepts at the end of the concept design cycle, where feedback was received on the validity and feasibility of the concepts, technical solution, and business models. The function of these interviews was to further solidify the trajectory and ensure that the adoptability of the concepts in theory is as high as possible. But also, for further research and concept design cycles to iterate on.

3.3.5 Observations in the local environment

Field research in the local environment was conducted in different parts of Stockholm. The purpose was threefold: firstly, to perform short *street interviews* with employees at various stores and restaurants that could potentially be future users of an intermediate storage. The purpose of these interviews was to gain an understanding of the problems and needs of that market segment in terms of their goods deliveries, and whether these could be addressed through introduction of an intermediate storage.

The second purpose of the observations concerned a later stage in the project when concept ideas were being developed. The purpose was to investigate local environments to determine whether there was any geographical space available to place the storages in different parts of Stockholm, which was crucial for assessing the feasibility of the concepts.

Lastly, the third purpose was to investigate how the problems and needs varied for different goods recipients in terms of goods deliveries due to how traffic and street conditions varied between different parts of Stockholm. This was to later confirm the need for more off-peak deliveries, as well as to assess the severity of the problem of heavy goods deliveries during the daytime.

3.4 Data analysis

This section provides an explanation of the different methods that were used to analyse some of the gathered data.

3.4.1 Interview summaries

After each interview, a meeting was held to analyse the interview and reflect on what the respondents had said and how it could be linked to the project. As an aid during the analysis, the autogenerated transcripts and recordings from Microsoft Teams were used to ensure that all analysis was based on the correct interpretation of what was said. Documentation was

conducted where the most important content from the interview was summarised, and quotes from the respondents that confirmed the analyses were written down.

3.4.2 Continuous feedback validation

Feedback was given from the industrial supervisor and the academic supervisor continuously throughout the project at a regular basis. This ensured that the project went in the expected direction from both supervisors and that the project did not stop for too long when obstacles emerged. During the supervision meetings, notes were taken so that information would not be lost and could be looked back on afterwards. Straight after the meetings, discussions were held where the feedback was analysed and addressed in three ways: what value the feedback provided, what from the feedback could be used practically and in what way, and what was the next step forward.

3.5 Concept design

This section presents the different steps involved in designing the concepts during the project. The concept design was divided into three concept iterations with different focuses, with the purpose of developing more elaborate concepts at each iteration. Continuous feedback was provided by various individuals which was essential to revise and improve the concepts throughout the project.

3.5.1 First iteration of concepts - technical solution focus

The focus of this iteration was entirely on creating a context for the project and therefore only developing technical solutions and not look at the business aspect. Based on data from the initial research phase of the project, the first iteration of concepts was developed, resulting in three distinct technical solutions that differed significantly from one another.

Individual brainstorming

The development of the first set of concepts began with an individual brainstorming process conducted by the project members over a period of six days. A set of criteria was established for the project members to consider during the brainstorming process. First, a general description of the technical solution was supposed to be included. Secondly, a mapping of the stakeholders involved in the logistics chain, as well as a description of the potential value and revenue they could get from being involved was also supposed to be included. Each individual was supposed to create three to four concepts, which were intended to be kept relatively simple rather than fully developed. The purpose of starting with individual brainstorming was to foster creativity and independent thinking, in order to minimise the risk of the project members being influenced by each other.

Collaborative brainstorming workshop

After the individual brainstorming, a collaborative brainstorming workshop was held between the project members to further develop the initial concepts. The workshop began with a presentation of the respective ideas and provided feedback. The concepts were compared to identify differences and similarities, and then they were evaluated based on which aspects were considered to have the greatest potential. Based on these evaluations, parts from different concepts were combined, resulting in a second version comprising three new drafts. Simple visualisations were created on a whiteboard to enhance clarity.

Finalising formulation and design of initial concepts

Once the workshop had ended and the concepts had been refined, more detailed descriptions and visualisations were produced. A PowerPoint presentation was also prepared to present the concepts at the final stage of the first concept development process.

First concept demo for the reference company's Innovation Office

The final step in the first iteration of concept development was to present the concepts to employees at the reference company's Innovation Office. This was an interactive session in which the audience asked questions and provided continuous feedback. The input was then evaluated to determine what changes needed to be made to the concepts, as well as what additional actions were required in the innovation process to ensure the project's continued progress.

3.5.2 Second iteration of concepts - technical solution and problem focus

The focus of the second concept iteration was to further develop the technical solutions based on feedback from the first concept demo. A mapping of identified problems and transactions along a logistics chain was created with the reasoning that these factors needed to be considered to further develop the technical solutions. The activities were carried out through an iterative process where they were performed in parallel.

Problem analysis concept solution in a logistics chain

Following feedback from the first concept demo, a brainstorming workshop was conducted where a logistics chain was defined. Relevant problems were specified for the different stakeholders in the logistics chain that could potentially be solved by introducing off-peak deliveries and intermediate storage. The result of this activity was used in order to develop the concepts further by assessing each problem, how each concept solved them, and in what way.

To map out the problems in detail, they were first identified, then why they exist, in order to understand how big the problems were. The problem owners and the actors who were affected by the problems were also identified in order to later assess potential ownership of the solutions. Parts of the problem analysis were inserted into a PowerPoint presentation.

When the problem analysis was made, the concepts were initiated in their formulation along with how each concept was thought to solve each problem, in order to assess their respective value proposition.

Collaborative brainstorming workshop

In combination with the problem analysis, a collaborative brainstorming workshop was done where the formulation and creation of the second iteration of concepts was made. It involved making changes to the concepts from the first iteration by selecting the most interesting parts from the first iteration, based on the feedback, and building on that.

Transaction analysis in a logistics chain

The transaction analysis was done for each concept's new logistics chain through a collaborative brainstorming process in a workshop session. The function of this activity was to assess what cost and revenue streams had to be identified, what figures had to be found through more research, as well as what had to be calculated moving forward, in order to later determine business model viability.

Finalising formulation and design of the second iteration of concepts

The formulation and design of the concepts were done by creating illustrations and descriptions for the ideas in each technical solution. All the concept formulations were inserted into a PowerPoint presentation in a preparation for the second concept demo session.

Second concept demo session for the reference company's Innovation Office

The second concept demo included a presentation of the problem identified in a current logistics chain and all of the concept variants, along with the proposed new logistics chain with each concept, how and where the solutions to the problems occur, as well as identified transactions. During the demo, time was allocated to individual feedback given using the tool, Mural. Although some feedback was also given continuously verbally. The demo was held for the reference company's innovation office, and in addition an external individual within the reference company with expertise in business models. The main purpose was to receive valuable information on the concepts from the reference company's perspective, and also an assessment of the concept's strengths and weaknesses.

Individual follow-up feedback sessions

To complement the feedback received during the concept demo session, follow-up meetings were held with some of the participants who wanted to complement their feedback and which the project members had follow-up questions with. These meetings were positive in that the participants had more time to assess the concepts presented in the concept demo, which in turn led to deeper analyses.

The feedback given during the demo and at the follow-up sessions were processed by analysing in what way the feedback would be applied in the project moving forward.

3.5.3 Third iteration of concepts – business model focus

In the third iteration of concepts, the concepts from the second iteration were developed further with more clear formulations of the technical solutions, the business models, and financial calculations, along with new sketches and AI-visualisations. This concluded the concept development, and the third iteration of concepts were then presented at the final presentation for the reference company as the final concepts.

Customer segmentation

A customer segmentation was conducted for one of the early identified market segments. The decision to do this segmentation was based on feedback from the second concept demo, where one of the participants pointed out that the initial segment was too broad, and that the actors involved had different problems, needs, and values. Therefore, it was necessary to separate the different actors in order to make more accurate estimations of those aspects, as well as to evaluate the validity of each concept.

Second problem analysis and concept evaluation of the new customer segments

A new problem analysis was conducted based on the first analysis in the second concept iteration, with the difference that the new one only included the new customer segments that were identified after the segmentation. The reason for analysing only these actors was that they were identified as one of the key actors in the logistics chain and might be the ones who would benefit the most from the new concepts. Therefore, a deeper and more focused problem

analysis was crucial. The analysis assessed the severity of the problems for each segment, as well as how well the concepts could both prevent and address the problems. The reasoning was to understand the severity of the problems for each segment, but also which segment was most in need of a solution for their problems. This in turn was used to assess which segment would be the most attractive customer.

Formulation of involved actors' roles

Roles were defined for every stakeholder that were involved for every business model according to the roles identified and described in Section 4.1.3. They are commonly used in transport ecosystems and were therefore deemed suitable to use for this scenario. It also provided clarity by assigning specific roles this way.

Cost and revenue analysis for the reference company and all customer segments

The different types of costs and revenue streams that would arise in each business model and technical solution for the reference company and their potential customers were identified in a cost and revenue analysis.

Finalising formulation and design of the third iteration of concepts

When all of the business models were completed, they were evaluated, and the least probable combinations of technical solution and business model were excluded. The final concepts were deducted from the formulated business models and an evaluation of their feasibility. The final concepts were the formulation of the most probable business models and technical solutions to be realisable in practice and possible to offer in the reference company's future product portfolio. The data from the cost and revenue analysis were summarised and compiled into a rough estimate of the significant costs of each concept. The calculation for the costs for the reference company's customers were based on how much they would have to pay in order for the reference company to earn a return of investment (ROI) after three years, which was the selected time period for all models.

3.5.4 Assumptions

Some assumptions in different areas of the project were made which were based on the collected data on which qualified guesses on the lack of data could be drawn, which constituted the assumptions. In addition, some were solely based on the project members own knowledge, opinions, and logical thinking.

Some financially related assumptions were made in order to make estimations and produce figures on projected financial feasibility of the technical solutions and business models in the concepts. Other assumptions were made when the concepts were developed regarding the assessment of them.

4. Results from research phase

This chapter covers the empirical results connected to the research phase of the project, including desktop research, interview sampling, and findings from the research interviews.

4.1 Desktop research on goods deliveries in urban environments

This section presents desktop research on goods deliveries in urban environments, off-peak deliveries, and stakeholders in the ecosystem of goods deliveries in urban environments.

4.1.1 Goods deliveries in urban environments

Urban freight is increasing and is expected to keep doing so due to continued globalisation and urbanisation, higher incomes, and the increase of e-commerce (Giuliano, 2023). In the field of sustainable urban freight transport, Verlinde and Machris (2016) argue that the objectives of the public and commercial stakeholders often conflict, which complicates the development of solutions that would receive support from all stakeholders. However, an innovation system involving all actors can be mutually reinforcing compared to if they were to approach the topic separately. The authors also state that co-operation between different stakeholders when developing and implementing new solutions in urban freight has been an ongoing topic. The idea is that service levels will increase, and costs will decrease through consolidation of goods. It has been stated by several in previous research and projects that partnerships between government and one or more private sector companies with a public-private understanding, are required for ensuring successful consolidation schemes in the long-run.

4.1.2 Off-peak deliveries

Off-peak deliveries are not yet commercially implemented broadly as an overall infrastructure in a large urban area due to the logistical and economic challenges with all of the involved stakeholders and their competing priorities. The challenges have been getting all actors simultaneously on board with the shift that has to be made and how to create a business model that benefits all parties. There have though been many successful pilot projects in different areas of the world demonstrating the upsides and opportunities of this transport system, such as increased transport efficiency and reduced emissions. Some of the successful pilot projects have resulted in off-peak deliveries being partially implemented in the city's plan (Sánchez-Díaz et al., 2016). An example of a pilot project conducted in Sweden was *Smart deliveries* carried out in Södertörn, south of Stockholm. The results were positive, and all units wanted to continue with Smart deliveries as this type of off-peak delivery made planning easier. Everyone appreciated that the goods were in place when the working day started (Closer, n.d.).

4.1.3 Stakeholders in the ecosystem of goods deliveries in urban environments

This section consists of a stakeholder mapping of the actors that are usually involved in the ecosystem of goods deliveries in urban environments. There are many different categories and names mentioned in the literature. Those presented in this report are some of the most common ones used in general mappings of ecosystem stakeholders, rather than linked to urban goods deliveries specifically. However, the stakeholders specific to that sector can fall into the same categories.

Customers
Customers are the goods recipients, but not always the end consumer (Fadwa et al., 2023). Customers are the ones that generate demands of the city logistics system. The customers range from all the firms, organisations, institutions, and private citizens who either ship, or receive goods through the system (Crainic et al. 2023).

Suppliers
There are many stakeholders that can fit under the term *suppliers.* In general, Fadwa et al. (2023) argue that they are stakeholders that are directly involved in urban transportation, which could be actors such as, *component suppliers*, *manufacturers*, *carriers*, and *retailers*, who usually are private companies. Crainic et al. (2023) uses the term carriers for the providers of the transport and warehousing resources and services the which makes the supply side of a city logistics system. Fadwa et al. (2023) continues by stating that private companies involved in goods deliveries seek to be efficient and cost-effective to meet the demands of the end customers by providing short delivery times.

Orchestrators
The orchestrator is considered to be one of the most important actors within an ecosystem (Lingens et al., 2022). It is the central organisation that advocates for the ecosystem's value proposition to other actors within and outside the ecosystem. The orchestrator is crucial to the health and stability of an ecosystem, as it facilitates the connectivity of complementors and users through the orchestrator's ability to shape the ecosystem's goals, identity, and design (Thomas & Ritala, 2021). Usually, the leadership role is attributed to an orchestrator, which is a company in an innovation ecosystem; however, in knowledge and entrepreneurship ecosystems, it is common for the leadership role to be taken by a facilitating or supporting actor, such as a university, a research organisation, the government, or an independent management organisation (Hoffmann et al., 2022).

The orchestrator is especially important in emerging ecosystems, as it needs to reassure the participants in the new ecosystem that value will be co-created between them, and it needs to convince users and other societal stakeholders of the viability of the ecosystem (Thomas & Ritala, 2021). They also need to establish the roles of the other actors within the emerging ecosystem and coordinate their interactions. These activities are particularly important in the beginning since there is a lack of structure in the interconnections between actors in an emerging ecosystem (Dedehayir et al., 2018).

Complementors
Complementors are actors in the ecosystem whose role includes providing complementary products, services, or inputs that contribute to the value proposition of the ecosystem. Complementors are crucial for the legitimacy of the ecosystem to emerge, as value propositions cannot be formulated or delivered without their participation. Only through the participation of the complementors in the ecosystem, legitimacy is achieved as the availability of inputs is crucial for the ecosystem value proposition (Thomas & Ritala, 2021).

Regulatory authorities
The government consists of people who has the authority to govern a country (Fadwa et al., 2023). The government is a key actor with the power to bring together actors from different

sectors and regulate and finance their activities. In this way, they can play an important role in introducing innovations at the societal level (Moore & Hartley, 2008). Flynn (2007) agrees by stating that governments play an important role in creating the economic conditions for growth and technological development. In many cases, they have systematic policies to encourage and subsidise innovation. A regulator specifically, are an entity that are responsible for ensuring efficient management of goods transportation and vehicles (Fadwa et al., 2023).

Public authorities' aim for urban transport is to minimise negative impacts that could come with transportation, to increase the attractiveness of the city for its citizens (Fadwa et al. 2023). Public authorities, which are the decision-makers in a city, are responsible for the state of cities and the quality of life of its citizens. Therefore, they play a fundamental role in the design of urban flows and should, when making decisions, take into account the needs of the different stakeholder groups involved in urban freight transport (Przybylska et al., 2023). Ecosystems can be legitimised by regulators by setting and enforcing standards that underline the new reality of the ecosystem and its role in society (Thomas & Ritala, 2021). There are three pillars that summarises the enabling of sustainable urban freight transport that regulators, as well as other relevant stakeholders, need to take into account. Firstly, *economic*, which is concerned with reducing costs of transportation. Second, *environmental*, which is concerned with the reduction of pollution, including noise. Lastly, *social*, which is concerned with increasing traffic safety and accessibility to transport and therefore to goods (Przybylska et al., 2023).

Indirect stakeholders

Indirect stakeholders are not actively participating in the city logistics system, but they are impacted parties who in some capacity influence freight-related activities. They include citizens, lobbying groups, real estate actors, and other associations. Indirect stakeholders can both get the benefits from having goods delivered or removed, but they might also get impacted negatively due to freight-related traffic and activities (Crainic et al. 2023). Citizens' priorities lite in safety and mobility within the urban environment that are unrestricted (Fadwa et al. 2023).

4.2 Desktop research on storage of goods in an urban environment

This section covers intermediate storage, other solutions of storage and storage of refrigerated goods in urban environments.

4.2.1 Market analysis on temporary intermediate storage

The usage and evolution of warehouse and distribution facilities has been increasing. The purpose is to move the goods from production to sale as quickly as possible; however, the distribution systems have been complicated by the increase in e-commerce and the need for shorter delivery-times. The distribution centres must be close to the customers to provide short delivery-times, which increases upfront and operation costs of the distribution facilities. The demand for large-scale facilities requires cheap land, which is more available outside cities. The result is an emerging distribution network where goods are shipped to large consolidation centres, which are then delivered to smaller hubs closer to the city (Giuliano, 2023).

The focus area of the market analysis on previous and present similar solutions to temporary intermediate storage was the Swedish market and European markets. The solutions are existing facilities where goods are positioned temporarily for the purpose of being consolidated with

other goods and transported to the recipient in a more efficient way. The presented solutions fall into the categories *hubs* – small enough to be positioned with close proximity to one another, or *consolidation centres* – large centralised distribution in need of large areas and therefore not suitable to be located in urban cities.

Swedish market

The Swedish market is highly focused on Business to Consumer (B2C) solutions in the area of temporary intermediate storage solutions for goods deliveries. No large-scale Business to Business (B2B) infrastructural solution in urban areas in the form of hubs was identified.

Instabee is a company which specifies in delivering parcels, with the main driving factor of making deliveries more sustainable. They own three companies where two of them, Instabox and Budbee, is based and operated in Sweden which specifies in delivering parcels to a hub which is located close to their customers' homes. The last-mile delivery (LMD) and personal information identification process is made by the receiver themselves instead of the old system where parcels are collected at a postal service point. Their hubs are positioned inside buildings, stores, or malls, making them accessible at limited times (Instabee, 2025).

iBoxen is another influential parcel storage competitor to intermediate storage. The company aims to be a neutral infrastructural enabler by being open for use by any logistics company (iBoxen, 2025). They also, similar to Instabox and Budbee, aim to reach customers close to their homes, erasing queues and interactions associated with traditional postal service points. Their unique value offering is that their storage hubs are located outside and therefore accessible to the logistics companies and recipients at all times, and not bound to the opening hours of a store or a mall.

Willys's concept *Hämta* is meant to store edible products, with an intermediate storage hub that can keep a steady cool temperature. This is meant to save the customers' time as well as potentially reduce their total travelled distance. Axfood has communicated an ambition to increase the number of these intermediate storage hubs for their other grocery store brands and therefore long-term create a new infrastructure for collecting food (Svensk Handel, 2021).

Urban Services is the only identified company which has a consolidation centre for B2B purposes in Arenastaden close to Stockholm where there is an influx of deliveries of goods and an outflux of waste. Their goal is to reduce transports by 70% by enabling joint loading and a common logistics flux with the centre (Urban Services, 2025).

European markets

The European markets, in countries such as Germany and the Netherlands, have come further with trying to implement temporary storage as a B2B solution.

Shurgard offers businesses in the Netherlands storage spaces in order to build their own distribution networks, and have their goods stored closer to their customers (Shurgard, 2025). These storage spaces give companies faster deliveries to their customers, reduces transportation costs, and enhances customer satisfaction. The flexibility of these units of storage allows companies to optimise their delivery logistics, adapting it to varying demand.

DPD is a company in the courier industry that offers package delivery services. The company is based in France but operates worldwide. They provide companies and private parcel recipients with an option to collect their packages at their urban depots located in city centres facilitating last-mile deliveries by bicycle or by foot (DPD, 2025). These urban depots only hold single parcels and not pallets of goods. In addition to these urban depots, they also have larger consolidation centres outside of cities to facilitate more efficient deliveries.

Storebox, similar to Instabee, is another parcel collection hub for end customers. They say that it is a situation where both parties benefit, both suppliers and customers. Route optimisation is a time and resource saver for suppliers while convenience with flexible ordering and collection of parcels is appreciated by customers. The global furniture and interior producer IKEA have partnered with Storebox to offer its customers valuable flexibility for parcel delivery in rural and urban areas (Storebox, 2025).

4.2.3 Storage of refrigerated goods

Perishable products require a precisely temperature-controlled atmosphere along the entire supply chain, from production to consumer. Therefore, refrigerated transport and storage are two crucial factors to prevent the quality of the product from declining (Shashi et al., 2020). According to Ali et al. (2018), a critical risk that can arise in the cold chain logistics of perishable products is a decline in product quality due to delivery delays, which can be prevented by efficient transport. This can maintain the value and saleability of perishable products, which is in line with Shashi's et al. (2020) statement. Pajic´ et al. (2024) adds that strict temperature control and monitoring during both transport and storage is of extreme importance. Furthermore, the authors claim that companies often rely on packaging for transport and overlook the importance of monitoring.

There are a few ways to achieve regulated temperature conditions during goods transportation. Pajic´ et al. (2024) argue that both active and passive systems are necessary. Active systems ensure correct temperature in insulated casings, such as cold storage facilities, temperature-regulated trucks, and refrigeration units. Passive systems are mechanisms that uphold correct temperature within insulated casings, either with or without thermostat regulation. This is achieved by a certain quantity of refrigerants, such as frozen gel packs, dry ice, or other similar materials.

4.3 Interview sampling

This section presents the sampling of organisations and individuals that were used for interviews.

4.3.1 Sampling of organisations for interviews

The interviews were conducted with respondents from eight different types of organisations and were divided into eight different categories: *truck manufacturing*, *grocery stores*, *cafés*, *restaurants*, *logistics companies*, *real estate companies*, *regulatory authorities*, and *universities/research centres*. Each category contained a variety of companies and organisations, and had a differing number of respondents, resulting in a total of 29 interviews.

The categories represented entities that either had a stake in the logistics chain or had knowledge of system innovation. All respondents were therefore able to contribute with

valuable insights to the project from different perspectives. Which organisational categories to include was determined during the research phase. However, which specific companies, organisations, and respondents were decided throughout the entire project, as the purpose of the interviews varied during the course of the project. Private companies that have a financial interest and that could be a potential customer are anonymous in order not to be traced, as well as the universities and research centres. The other organisations, which either have no financial interest or other economic impact, are public and mentioned by name. All respondents from all organisations are completely anonymous and not mentioned by name. See Table 1 for an overview of the interview sampling.

Table 1. Interview sampling

Organisational categories	Organisations	Nr. of interviews
Truck manufacturer	The reference company	3
Grocery stores	Grocery store A Grocery store B Grocery store C	3
Cafés	Café A Café B Café C Café D	4
Restaurants	Restaurant A Restaurant B Restaurant C Restaurant D	4
Logistics companies	Logistics company A Logistics company B Logistics company C Logistics company D Logistics company E	5
Real estate companies	Real estate company A Real estate company B	2
Regulatory authorities	Municipality A The Swedish Police Authority The Swedish transport administration The Swedish Transport Workers' Union	5
Universities/Research centres	University A Research centre A Research centre B	3

4.3.2 Motivation for the sampling of companies and organisations

Each organisational category served their own purpose for why they were relevant to be interviewed and for what reason they were selected.

Truck manufacturing

There were two main reasons for interviewing employees at the reference company. First, the reference company had valuable expertise in both logistics and system innovation. Secondly, as parts of the results from the project were for the reference company, it was necessary to conduct all types of interviews with them with several respondents to gain as much insight as possible so that the best necessary conditions could be obtained.

Goods recipients: grocery stores, cafés, and restaurants

These categories were essential to include in the interviews, as these businesses represent the end customers in the logistics chain and would be directly affected by potential changes. The selected grocery stores, cafés, and restaurants were chosen semi-randomly during the field observations in the innovation phase. They were not pre-selected before the trips; instead, the decision to approach each business was made on site, based on whether they appeared to be potential future users or customers of an intermediate storage. The decision was based on their specific location, the surrounding environment, the nature of their business, and whether the staff seemed available to talk or if they appeared too busy with customers.

Interviews were also conducted with respondents from these categories during the validation phase. A more careful selection was made for these interviews, with the ambition being to have at least one from each category in the same areas where the field observations were made. This was to be able to present the concepts that were most suitable for each company.

Logistics companies

Logistics companies was another essential category for interviews, as they are a key actor in the logistics chain, acting as the link between the suppliers and the goods recipients. They were identified as potential customers and users of an intermediate storage, which is why they were selected for interviews.

Six companies with different business models were interviewed. Some specialised in long-distance goods transport, while others performed LMD. The types of goods they handled also varied: some focused on refrigerated goods for stores and restaurants, while others handled home deliveries of smaller packages. One company did not operate any logistics or transportation of its own, but instead provided small-scale intermediate storages used by both their customers who were transport companies and by the transport companies' customers who were private individuals. This variety of logistics companies was selected in order to gain insights from several parts of the logistics chain, as well as to understand how different types of goods influenced the delivery process.

Real estate companies

During the early stages of the research phase, potential issues were identified regarding the placement of the storages, especially in relation to municipal land. As a result, it was decided to investigate the possibility of placing storages on private property. Therefore, real estate companies were selected for interviews in order to gather data on this topic, but also to investigate their willingness to take an active role in the logistics chain and thus also in the business model.

Regulatory authorities

The regulatory authorities interviewed in this project were municipality A, the Swedish transport administration, the Swedish Police Authority, and the Swedish Transport Workers' Union. It was decided early in the project that these governmental authorities would be relevant to include.

Municipality A was a key stakeholder in the project and was therefore interviewed for several reasons. Firstly, the municipality had a strong need to reduce heavy goods traffic in the city during daytime hours, as it is responsible for the city's environment and the health of its citizens. This made it important to investigate and understand the problem from their perspective. In order to develop viable concepts, it was also necessary to learn about the opportunities and constraints defined by the municipality, so that these could be taken into consideration during the concept development.

The Swedish transport administration was selected due to its work on innovations related to transport and urban infrastructure, as well as its influence on the adoption of such innovations by relevant stakeholders.

The Swedish Police Authority was selected due to the need for insights regarding the security aspect and risk of criminal activity in connection with new system innovations in general and specifically to goods transportation.

The reason for including unions were to gain insight into how changes in the logistics chain, and the introduction of intermediate storage might affect the involved stakeholders from a union's perspective, especially in terms of safety for both the drivers and the goods recipients. The selected union respondent was the Swedish Transport Workers' Union, since they could provide both general insights from a union's perspective, and knowledge related to transport, which was highly relevant for this project.

Universities and research centres

The reason for interviewing universities and research centres was to gather theoretical and general knowledge about critical factors that influence the development and implementation of new system innovations, and specifically, what affects the adoption of such innovations by the involved stakeholders.

Three institutions were interviewed: university A, research centre A, and research centre B. These were selected due to their expertise in different relevant areas. The respondent from university A specialised in parcel logistics, research centre A focused on the introduction of different types of innovations, and research centre B were conducting research on logistics and transportation. These were all valuable perspectives to address in the project, which is why they were included.

4.4 Findings from research interviews

This section presents the main findings from the research interviews regarding the five most discussed topics: *system innovation*, *goods deliveries*, *off-peak deliveries*, *intermediate storage*, and *ownership and roles*. The organisational categories included in those in those interviews were: *truck manufacturer*, *logistics companies*, *regulatory authorities,* and *universities/research centres*, and in total, nine research interviews were conducted.

4.4.1 System innovation and adoption

In the interview with the respondent from research centre A, it was highlighted that system innovation is a foundational condition for all other innovation, and that it enables more technical oriented innovation. According to one respondent from the reference company, there has evolved new ecosystems since the 2000s which have brought cooperation challenges with regards to innovation, mainly involving unclear roles and responsibilities, profit distribution, problem ownership, and intellectual property rights. Many projects ultimately fail to be implemented in practice because these challenges remain unresolved. The respondent from the reference company also emphasised the importance of ensuring that all decision-makers fully understand the system as a whole. Creating a sense of security or a *comfort zone* for involved actors was described as crucial for enabling effective collaboration. In addition, the integration of partners' business models was highlighted as a vital factor when pursuing system innovation. By aligning and leveraging the key resources of various actors, network efficiency can be improved, ultimately increasing the likelihood of a successful outcome.

Regarding innovation adoption, it was mentioned by the respondent from research centre A that focus should on achieving acceptance should be with early adopters to ensure long-term success, because eventually late adopters will still need to adapt and accept the innovation, especially a system innovation. The respondent from research centre B added that adoption, acceptance, and legitimacy, can be better achieved by having citizen dialogue regarding issues that can have a high impact on that stakeholder group. General adoption was said to be facilitated by having citizen dialogues by the respondents from municipality A as well. They communicate directly to and with citizens of the municipality to get their perspective on certain issues, such that they do not make decisions which could upset society. Another important factor that is important to notice regarding adoption, discussed by the respondent from research centre A, was the underestimation of people's unwillingness to change habits and try something new. People might not even want to admit that they have needs which could be solved by innovations. This phenomenon makes it hard to introduce and ensure adoption of radically new innovations. The first step in ensuring acceptance was according to the respondent:

> *"Go out and talk to people you believe to be a potential customer. If they say 'What...?' they are wrong. Then they are too far away on the adoption curve. They are late laggers. They will never buy this either way until everyone else has bought it. Drop them. Do not waste time on those who do not want it."* (Translated by the authors, anonymous participant, personal communication, March 20, 2025)

The respondent from the Swedish Transport Administration believed the best possible way to achieve adoption of system shifts is to introduce the solutions in small volumes and on a small scale in a single location. An example of an attempt at adoption of a new system shift mentioned by the respondent was where they had identified a road north of Gävle that was very dangerous where many fatal accidents occurred. The road had been identified as a crisis area where it was critical to find a solution. Therefore, they introduced a centre barrier on the road that would prevent collisions. This type of solution had never been put in place before which, after its introduction, drastically reduced the number of accidents and a few years later, the solution had been scaled up considerably with more roads having central barriers. The introduction was met with scepticism initially, but acceptance was eventually gained.

> *"It's just that a lot of people died before the railing was put up and when the railing had been up for a year, no one had died. There was no way to argue with that. This was simply a good idea. Then it took a few years, then almost all roads were centre-separated, and we had saved about 200 lives."* (Translated by the authors, anonymous participant, personal communication, June 18, 2025)

Related to that example of adoption taking time, the respondents from municipality A also mentioned the "*silent positive majority*" and that the people they receive viewpoints from mainly are negative people. As change usually is met with negativity in the beginning, they mentioned that it takes time for adoption to happen. An important factor was said to be able to counter worries initially with answers on how the worries will be handled.

4.4.2 Goods deliveries

Regarding current goods delivery practices, the respondent from logistics company C noted that their operations were largely dependent on the opening hours of their customers, the goods recipients. This reliance was mentioned as a constraint that limits the logistics company's flexibility and ability to make their processes more efficient. Looking ahead, one improvement potentially increasing efficiency mentioned by the respondent from logistics company C was temporal consolidation of goods, meaning to spread out deliveries over time. However, respondents from municipality A pointed out that there were currently no incentives for such consolidation. Goods recipients today benefit from fast, personalised deliveries, which discourages any shift involving slower deliveries even if the benefits from the logistics, economic, and sustainable perspective are big. The respondent from research centre B pointed out the same issue and emphasised that strict delivery time slots imposed by recipients hinder the ability to load trucks with a maximum degree of filling. It was argued that greater flexibility from goods recipients would be essential for increasing truck fill rates, which is a key condition for achieving consolidation and more efficient delivery operations. The primary benefits of consolidation and off-peak deliveries was said to favour logistics companies and the city more than the recipients. This model was said to potentially be used here as well to incentivise the goods recipients.

4.4.3 Off-peak deliveries

The interview with the respondent from logistics company A revealed a strong willingness to expand off-peak deliveries, which they viewed as highly beneficial to their operations as they could possibly reduce the amount of vehicles and lower the total system cost. The respondent from logistics company B similarly highlighted that flexible delivery times, being able to deliver at any hour, improve the efficiency and return on investment for costly assets like trucks. From the municipality's perspective, the respondents from municipality A emphasised that the concept of large-scale off-peak deliveries have the benefit of increasing the utilisation of urban space and infrastructure across more hours of the day. However, they noted that for off-peak deliveries to be viable, vehicles must operate with minimal noise to avoid disturbing residents. The respondents from municipality A also pointed out that from their experience delivery personnel who work night shifts were generally reluctant to return to daytime schedules, largely due to the advantages of night-time deliveries, such as reduced traffic and more efficient delivery conditions. This positivity towards off-peak was confirmed by the respondent from research centre B who noted that a previous pilot project involving off-peak deliveries of goods to a school had been highly successful and that all participants involved,

the logistics company and goods recipient, in the project had expressed an interest in continuing the initiative.

A major challenge of convincing goods recipients to accept deliveries earlier than their current receiving times was expressed by the respondent from logistics company A, confirming the operational constraints experienced today by logistics company C. The respondent also noted that shifting to off-peak deliveries represented a significant change for recipients, especially due to the added cost of scheduling staff during nighttime. Despite this barrier, those goods recipients who have attempted off-peak delivery practices were said to have experienced improvements, particularly in the efficiency of their goods receiving processes.

4.4.4 Intermediate storage

The respondent from research centre B highlighted a key operational challenge if multiple logistics companies would be involved. Specifically, transshipment of goods to a third party, other than the intended recipient, could be problematic, as each logistics company remains responsible for ensuring that deliveries are made on time and without damage. This issue of accountability was said to create hesitation around shared use of intermediate storage facilities.

The respondent from logistics company A expressed an interest in delivering to intermediate storages during off-peak hours, provided that the final recipients are responsible for collecting their goods. This model would allow for more flexible delivery scheduling for them while not involving a third-party logistics company as noted by research centre B. The respondent from logistics company B said that intermediate storage in the form of parcel boxes achieved flexibility for the parcel collector through the use of both indoor and outdoor parcel boxes, where they could collect their parcel at any time. A similar solution for business deliveries, they noted, could offer comparable benefits to logistics companies.

The respondent from university A addressed another practical concern connected to the handling of temperature-sensitive goods. Extended storage times, which off-peak deliveries and intermediate storage may inflict, pose risks for refrigerated goods, particularly if the storage facility lacks cooling capabilities. In such a scenario, it was said that specialised packaging would be required to maintain consistent temperature levels to protect the goods.

Regarding physical space for placement of storages, one respondent from the reference company noted that municipality A does not allow long-term placement of storage facilities in public areas, and that current solutions for temporary intermediate storage are still under evaluation and will eventually be removed. The respondents from municipality A confirmed that intermediate storages cannot be placed permanently on municipal land, as the city aims to preserve as much open space as possible for its citizens. The municipality's overarching goal was said to be prioritising the well-being of residents and to create attractive urban environments. This was described as especially challenging in areas such as Gamla Stan, where available space is extremely limited or entirely lacking, making the placement of intermediate storages particularly difficult. However, the respondent from the reference company mentioned that regulations vary between different municipalities, and that some are open to providing space for an intermediate storage on construction sites or other unused areas.

To address these limitations in areas where reluctancy of providing space is present, the respondent from research centre B suggested that property developers should proactively

consider off-peak delivery and intermediate storage needs when designing new buildings. Incorporating such infrastructure into future developments could be a key enabler for advancing these logistics concepts in urban areas.

4.4.5 Ownership and roles

The interview with the respondent from logistics company D revealed that they were open to sharing responsibilities within the logistics chain and that they were willing to handle LMD, provided that their role would be financially viable and include return of goods in the business model. From municipality A's previous experiences with project regarding goods deliveries, logistics companies have expressed challenges with collaboration between different logistics companies, as they had observed a general reluctance among logistics companies to partner with one another. Furthermore, the municipality emphasised the need for any proposed solution to support a competitive market, allowing various logistics actors to access and use shared infrastructure flexibly.

Regarding municipality A's potential role in a concept enabling off-peak deliveries, the respondents noted that their primary role in system innovation projects prior has typically been that of a coordinator, supporting other stakeholders rather than leading or owning infrastructure. Similarly, the respondent from research centre B expressed interest in taking on a coordinating role in future off-peak delivery initiatives, noting that as a non-profit organisation focused on societal well-being, they are well-positioned to facilitate collaboration among actors without any commercial interests.

5. Results from innovation phase

This chapter presents the results from the innovation phase, including the concept design, observations in the local environment, assumptions, and findings from the concept interviews. The concept design was done in three iterations with different focuses: *technical solution focus*, *problem and technical solution focus*, and *business model focus*. The concept design has to some extent been based on the methodology presented in Section 2.2, but it has been adapted to better fit the nature of the project and personal preferences.

5.1 First iteration of concepts - technical solution focus

This section presents the results of the first iteration of concepts, with a description of the developed concepts along with the results of the first concept demo. The first iteration of concepts provided a foundation for the trajectory of the concept design in the project.

5.1.1 Finalised concepts in the first iteration

The initial concepts varied greatly from each other. They included a technical solution which could be implemented today, along with aspects for with could be implemented in the future. Visualisations were made for every concept that can be viewed in Appendix A.

Concept 1: Utilisation of existing facilities

The idea was that instead of building and operating new facilities, existing spaces would be used for intermediate storage. The transporter would deliver the goods to the storage in the evening, which is located in empty or unused facilities that could be rented out and used as intermediate storage. The goods are then delivered to the customer by the same logistics company, using several smaller vehicles to avoid heavy traffic in the city.

Another variant was that an organisation that has a warehouse and rents out storage to private individuals could rent out part of its premises for intermediate storage, where the goods would be delivered at night and consolidated.

In a future perspective, there is potential to increase the use of cool basement spaces for intermediate storage of refrigerated goods. The development of new urban areas and buildings should provide accessibility so that these spaces can be used.

The challenge of making this a profitable solution for the customers will probably require that the solution allows for unmanned reception. Customers will not see enough value in off-peak deliveries if someone must be on site to receive the goods. Therefore, some kind of technical solution is needed to allow the goods to enter the store. There was also a challenge regarding the ownership of the service and the payment for the use of the intermediate storages.

Concept 2: Mobile micro hub gateway

The second concept was based on intermediate storage being located directly at the customer's premises and thus without consolidation, i.e. only private storage. The transporters would deliver the goods from the retailer to a mobile micro-hub that is directly adjacent to a loading door into the customer's building. The hub would have two doors with two separate locks, one entering the hub from the outside and another facing the loading door into the larger building. As the hub would be very small in size and is directly adjacent to the main building, it should

be possible to integrate refrigeration. There would be a docking mechanism between the truck and the hub so that the door to the hub is never open to the outside, to make it easier to maintain the cold chain, but also for security reasons against theft.

The limitation of this solution is that only customers who have the physical space required for the hub can use this solution. Then there may also be building permits or other restrictions that may prevent these from being placed, which is something that needs to be investigated further.

Concept 3: Autonomous trucks as a consolidation hub

The last idea was based on an intermediate storage inside a truck rather than in a stationary facility. Transporters would collect goods from the retailer and deliver them during the day to consolidation centres on the outskirts of the city. The consolidation centres would be equipped with small autonomous electric trucks manufactured by the reference company, with each centre accommodating around 2-4 trucks. The driver of the larger truck would transfer the goods to the smaller vehicles, where the refrigerated goods would be loaded into mobile coolers inside. When the small trucks are fully loaded, they remain in the consolidation centre until nighttime when they would drive into the city.

Around the city centre, municipality A would have allocated parking spaces for the autonomous trucks in suitable locations with high demand for consolidated goods. The parking spaces would be equipped with electric charging posts that would allow the trucks to be charged, but also to keep the coolers refrigerated. These parking spaces would be prohibited for the public to use during specified times at night, which would be municipality A's responsibility to enforce. Early in the morning, a last-mile logistics company would carry out the delivery from the truck to the end customer, bringing a cooler or pallet on specially adapted electric bicycles. There should be enough parking spaces well placed around the city to minimise the length of the last-mile trips, to allow the bicycles to make as many trips back and forth as needed to deliver all the goods in a short period of time. As soon as all deliveries have been made, the trucks would drive back to the consolidation centres and the parking space would be open to the public again. This means that the storages would only occupy space in the city for a limited time during the night and leave before the morning rush hour.

The challenges in this concept mainly lied in the ownership of the consolidation centres and which actor that would control the entire operation. Some benefits for The reference company would be the opportunity to expand their manufacturing of a new type of electric and autonomous trucks, and it would provide a safe environment with appropriate conditions for testing of the trucks as it is carried out at night when there is less movement on the streets. There is also the possibility that The reference company could be the owner of the operation of the consolidation hubs and sell them together with the trucks as a service. In this way, they would have more control and an even greater influence in the business.

5.1.2 Results of the first concept demo

Overall, the feedback from the audience was positive. They recognised the potential and benefits of all concepts, but also some obstacles and drawbacks. Concept 1 had the advantage of the utilisation of existing buildings, which removed the obstacle of placing new facilities in crowded areas. However, they considered it to be the least well thought-out and least innovative concept, with many problems regarding delivery and customer reception. Concept 2 was

considered to have the most potential as it would probably be the easiest concept to implement with the current infrastructure. Concept 3 was found to be very interesting and an idea with a lot of potential, mostly because the storage was mobile. However, the number of actors and the number of steps in the operation would make the concept difficult to implement in practice.

5.2 Second iteration of concepts - technical solution and problem focus

This section presents the results of the second iteration of concepts. The finalised concepts, problem analysis, cost and revenue analysis and the feedback results from the concept demo session are highlighted in this section.

5.2.1 Finalised concepts in the second iteration

The formulation and design of the second iteration of concepts resulted in two main concepts, *A* and *B*. The main difference of them was the placements of the storages, where concept A consisted of a free-standing storage, while concept B consisted of a storage standing against the goods recipient's outer wall in front of their loading door. Within concept A and B, two variants with more differences were formulated, focused on technical solutions of identified problems, as well as other factors that could bring value to the users. In total, four concept variants were formulated: *A1*, *A2*, *B1*, and *B2*.

Concept variant A1: Mobile free-standing consolidation storage

Concept variant A1 involved a mobile free-standing storage for consolidated goods to several goods recipients in separate sections. The idea was to have the storages be positioned close to the connected goods recipients so that the actor performing LMD can do it effectively at night with unmanned reception, or early in the morning with manned reception.

To clarify the idea of the placements of the storage units, a sketch of Södermalm was created, divided into hypothetical zones. Within the zones, placement areas were visualised that would be designated for short-term placement of the storage units. The idea was that each storage unit would remain in a given location for a certain period of time and then be moved to another location in the same zone. This rotation of storage units, where there would be a larger amount of placement areas in relation to the number of storage units within the same zone, would ensure that a storage unit would not be permanently located in a particular location on the municipality's land. The sketches illustrating concept A1, both of the zones and the storage, can be viewed in Appendix B.

Concept A2: Stationary free-standing consolidation storage

Concept A2 was a free-standing storage unit for consolidated goods to several goods recipients that would be stationary in the courtyard of a private property, within short walking distance of the recipients. The goods would be unloaded in the storage unit during off-peak hours in sections for each recipient, with the goods being available for the recipients to collect at any time during the day. The key feature of this concept was to allow for unmanned receipt of goods and for LMD to be carried out by the recipients, potentially lowering their delivery costs. The sketch illustrating concept A2 can be viewed in Appendix B.

Concept B1: Stationary non free-standing privately used storage

Concept B1 was a privately owned storage unit with a single goods recipient, which means that there was no consolidation in this concept, unlike concept variants A1 and A2. The transporter

would deliver the goods to the storage at off-peak hours with unmanned reception. Another key difference of concept variant B1 to A1 and A2 is that the storage would not be free-standing, but instead it would be located directly adjacent to the goods recipient's facility and their loading door, where the employees could access the goods easily through a backdoor of the storage. This would allow the recipient to collect their goods at whatever time that suits them. They would also not have to go out as they have to do in A2. The sketch illustrating concept variant B1 can be viewed in Appendix B.

Concept B2: Stationary non free-standing consolidation storage

Concept variant B2 was a combination of A2 and B1. It was intended to be similar to A2 in that several different recipients in the same building would use the storage unit. It would be located in front of one of the goods recipients' loading doors, the same as in B1, while the other recipients in the same building would have to go outside to collect their goods, as in concept variant A2. The goods recipients that would have the storage closest were called *the primary goods recipient* and the rest were called *the secondary goods recipients.* The sketch illustrating concept B2 can be viewed in Appendix B.

5.2.2 Problem analysis of the logistics chain that off-peak deliveries and intermediate storage have the potential to solve and their respective solutions

A workshop was held where a problem analysis was created, which resulted in 13 identified problems that exist today for different actors in a logistics chain. An identification of which actors that "*owned*" the problems (the ones that would be responsible to address and solve them), who was directly affected by the problems, and why the problems exist was also included in the table. The problems and the other aspects were identified through educated guesses based on findings from the research phase. The problems were placed along a visualisation of a logistics chain showing where they occur. A simplified version of table showing the problems, and the logistics chain can be seen in Figure 2, while the entire elaborate table can be seen in Appendix C.

Nr	Problem	Nr	Problem
1	High load in the transporter's consolidation centre	8	Incorrect delivery
2	Suboptimal route planning due to order requirements	9	Unsustainable strain on staff
3	Access restrictions while driving	10	Excessive/costly storage space
4	Lack of optimal loading space	11	Excessive/costly inventory
5	Damaged goods/Broken cold chain	12	Shortage of goods to sell
6	Delayed delivery to the recipient	13	Shortage of goods to purchase
7	Manual receiving of goods		

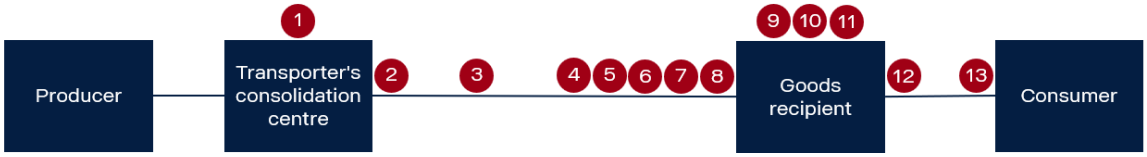


Figure 2. Identified problems and a mapping of them in a logistics chain.

Two new tables were made after the first one, one for concept A and concept B (the concept variants were grouped together), describing hypothesises of how they could solve each problem, i.e., how intermediate storage and off-peak deliveries could be a solution and in what

way. Most problems were solved the same way by concept A and B, but some differences occurred between the concept variants which were highlighted. Along with the hypothesises for the solutions of each concept, possible follow up problems were identified. The solutions concept A provided were hypothesised to, as a whole, generate more potential follow-up problems than concept B. These are presented in more detail along with a more elaborate evaluation of each concept's problem solution hypothesis, specific measures in the solution, actors potentially benefiting economically, and why they would benefit economically in Appendix D.

The new logistics chain that would be established with the introduction of the concepts was similar to how it looks today, but with the added intermediate storage between the transporter's consolidation centre and the goods recipient. In concept B2, an additional distinction between primary and secondary goods recipients was made. Problems were solved differently depending on the concepts and their respective variants as presented below.

Problem 1: High load in the transporter's consolidation centre

This problem occurs in the transporter's consolidation centre and can be caused by too high demand at a given time and a large number of orders and customers.

Concept A's solution to problem 1 was considered to be solved by the possibility of transporting the goods directly to the intermediate storage instead of burdening a consolidation centre, regardless of the variant. Concept B's solution to problem 1 would be unaffected as they were considered to be solved in the same way as today. The storages themselves would not provide a new solution to this problem.

Problem 2: Suboptimal route planning due to order requirements

This problem occurs during the transport of the goods between the transporter's consolidation centre and the goods recipient, which can be caused by large orders that may cause sub-optimal route planning and therefore sub-optimal resource utilisation for the transporter.

Concept A's solution to problem 2 was hypothesised to be solved by using fewer vehicles from the consolidation centre to perform large orders with, regardless of concept variant. Concept B's solution to problem 2 would be unaffected as they were considered to be solved in the same way as today. The storages themselves would not provide a new solution to this problem.

Problem 3: Access restrictions while driving

This problem occurs during the transport of the goods. Weather, traffic or a driver deviating from the planned delivery route are examples of access restrictions that can occur during the transport.

Concept A's solution to problem 3 would be solved by larger time margins due to off-peak deliveries. Concept B's solution to problem 3 was considered to be through the off-peak aspect making access restrictions less of a risk.

Problem 4: Lack of optimal loading space

This problem arises when the goods are unloaded to the goods recipient due to, for example, poorly adapted building architecture, or sub-optimal environmental factors.

Concept A's solution to problem 4 was considered to be solvable in different ways between A1 and A2. A1 would ensure a better possibility for the driver to deliver to a standardised storage interface while A2 would make the problem non-existent as the LMD would then be performed by the goods recipient themselves. Concept B's solution to problem 4 would be solved by a standardised unloading process that optimises the loading space.

Problem 5: Damaged goods/Broken cold chain

This problem can occur during transport, but also earlier or later in the logistics chain. The problem arises due to deficiencies in the handling of goods, where transhipments increase the risk of goods being damaged and thus unsaleable.

Concept A's solution to problem 5 would be through greater time margins to fix the problem thanks to the off-peak aspect. Concept B's solution to problem 5 would be for the driver to check the goods when delivering them, combined with the off-peak aspect providing a greater time margin to possibly address and replace the damaged goods.

Problem 6: Delayed delivery to the recipient

This problem occurs during transport and affects the goods recipients when they want to receive the goods at their desired time. The problem causes a chain reaction of delayed deliveries which is a lose-lose situation for the transporter and their customers.

Concept A's solution to problem 6 was considered to be off-peak deliveries, which would reduce the risk of delays. Concept B's solution to problem 6 was considered to be through the off-peak aspect and by having the goods in time in stock directly adjacent to the recipient.

Problem 7: Manual receiving of goods

This problem arises during the unloading process only when the recipient has to be present and can cause problems such as boom driving or when there is a need for assistance from the recipient.

Concept A's solution to problem 7 would be solved differently by A1 and A2. In A1, there would be no risk of boom driving if unmanned reception was ensured, while A2 made the problem non-existent for the carrier and the recipient as the pick-up time is determined by the recipient themselves. Concept B makes the problem non-existent as there is no requirement for assistance from the recipient or for them to be present, which means that there would be no risk of a boom drive.

Problem 8: Incorrect delivery

This problem occurs when goods are delivered to the wrong recipient. It can be caused by a misaddressed order or a human error by the driver. This is a time-consuming problem to solve.

Concept A's solution to problem 8 would be equally good in both concept variants through possible additional technical solutions, e.g. a tracking and tracing system for the goods, but also through larger time margins to solve wrong deliveries due to the off-peak aspect. Concept B's solution to problem 8 would be through greater time margins to resolve incorrect deliveries, due to the off-peak aspect.

Problem 9: Unsustainable strain on staff

This problem occurs for the goods recipients and is caused by problems earlier in the logistics chain, potentially leading to unscheduled staff deliveries and also rescheduling of the goods recipients' staff, which brings additional costs for the goods recipients.

Concept A's solution to problem 9 would be different for the concept variants. A1 provides more punctual deliveries which contribute to less risk of strain on staff, while A2 would both ensure punctual deliveries and provide an opportunity for the goods recipients to collect goods at their convenience. Concept B's solution to problem 9 would also be solved by more punctual deliveries and because the goods would already be in the storage when the recipents arrive, generating less time-related burden on their staff.

Problem 10: Excessive/costly storage space

This problem occurs for the goods recipients, which arises because they have an unnecessarily large and expensive warehouse space, which removes the possibility of larger store space, which means an unnecessary cost.

Concept A's solution to problem 10 was considered to be possible for A1 through the possibility of having buffer stock for holidays and high-season goods, only when the recipient of the goods would be in need of it, which means that they get better use of their warehouse space and can have larger store space. A2 would also ensure buffer stock that would be even more easily accessible. Concept B's solution to problem 10 would also be solved with the possibility of using the storage for as a buffer stock during high seasons when there would be an increased demand for certain types of goods. This would hopefully cover the cost of the storage space.

Problem 11: Excessive/costly inventory

This problem arises for goods recipients when they have expensive capital-intensive inventories which are unnecessarily tying up capital, potentially reducing the efficient use of capital.

Concept A's solution to problem 11 was considered to be possible for A1 through the possibility of having buffer stock for holidays and high-season goods, only when the recipient of the goods would be in need of it, which means that they get better use of their warehouse space and can have larger store space. A2 would also ensure buffer stock that would be even more easily accessible. Concept B's solution to problem 11 would also be solved with the possibility of using the storage for as a buffer stock during high seasons when there would be an increased demand for certain types of goods. This would hopefully cover the cost of the storage space.

Problem 12: Shortage of goods to sell

This problem arises in the relationship between the goods recipient and the consumer, from the recipient’s perspective. A shortage of goods to sell creates a negative customer experience for the consumers and lost revenue, both at the time of the purchase and potentially in the future if the consumer chooses not to return to the store. This is a problem that the goods recipients need to avoid as much as possible.

Concept A’s solution to problem 12 would be different for the concept variants. A1 provides more reliable and earlier deliveries, and A2 by having the availability of the storage reduces the risk of not having goods available for sale when the recipient needs them. Concept B’s

solution to problem 12 would be solved by the concept providing less risk of goods not being available for sale, since goods can be easily be collected from the storage and brought into the store when needed.

Problem 13: Shortage of goods to purchase

This problem also occurs at the relation between the goods recipient company and the consumer, from the consumers' perspective. A shortage of goods to purchase creates a bad customer experience and lost consumption of goods, both at the time of the purchase and potentially in the future if the consumer chooses not to return to the store.

Concept A's solution to problem 13 would be different for the concept variants. A1 provides more reliable and earlier deliveries, and A2 by having the availability of the storage reduces the risk of not having goods available for sale when the recipient needs them, which means that the goods will be available for purchase by the consumer. Concept B's solution to problem 13 would be solved by the concept providing less risk of goods not being available for purchse, since goods can be easily be collected from the storage and brought into the store by the goods recipients when needed.

5.2.4 Transaction analysis in the new logistics chain

A transaction analysis was made for each new logistics chain for their respective concept. The motivation of doing this was to get an understanding of how the new system costs would look like and how they compare to today. The visualisations of the transactions identified in each new logistics chain for each concept can be viewed in Appendix E.

The identified new transactions which would incur new costs for the actor paying for it were storage production/purchase and distribution, transportation of goods on off-peak hours, rotation of intermediate storages, placement of storages on either municipality's or on real estate companies' land, goods recipient's labour, and the total service cost for the customer.

5.2.5 Results of the second concept demo

Feedback was given during and after the concept demo session for the reference company's Innovation Office in the following categories: *The reference company's role, Concept evaluation,* and *General input moving forward.*

The reference company's role

Valuable input was received regarding the reference company's role in the system. The feedback showed that the highest potential is the reference company being in a coordinator role where their main responsibility is to handle contracts and partnerships in the entire system. Other feedback was for the reference company to partner with suppliers and/or logistics companies to offer the storage and off-peak delivery solution. It was mentioned by one of the participants that the final market offering in the concepts could be offered as a business opportunity for the reference company's existing customers. Their existing customers could gain new value-added services from the concepts and their connected business possibility, potentially increasing their revenue.

Concept evaluation

Overall, there was more confidence in concept B due to more costs and complex solutions connected to concept A. Risks with A1 specifically being very costly were mentioned by

several participants. Concept A was although mentioned to be a good initial starting point and a concept which has a potential to scale up concept B, as it would give a good starting point at evaluating needs and demands from goods recipients and transporters.

General input moving forward

Problem 10 was mentioned as interesting to look more into, and connected to this problem, evaluate flexibility in the concepts for goods recipients. A question to be considered was if the recipients would be able to choose when to have extra storage and the added expense along with that. It was highly suggested to look into Urban Services' business model regarding their consolidation centre. It was also recommended that the goods recipients market segment should be divided into smaller segments as it was too broad.

5.3 Third iteration of concepts – business model focus

This section covers the results of the final iteration of the concepts. The concepts were complemented with clear business models and calculations on the economic system feasibility.

5.3.1 Customer segmentation of the goods recipient market segment

A customer segmentation of the goods recipient market segment was carried out following feedback from the second concept demo, where one of the participants pointed out that the segment was too broad and would benefit from being split into smaller parts. It was divided into three smaller segments: *cafés*, *restaurants*, and *grocery stores*. The reason why these actors were chosen was because they all handle refrigerated and fresh goods, which is their main selling item, i.e. they all have similar type of needs for their deliveries. Unlike e.g. pharmacies which was an actor that was excluded as they sell medicines, which can indeed be refrigerated, but it is not one of their primary type of goods and thus they have other needs for their deliveries. The scope of the project needed to be kept narrow, and therefore this selection was made.

The customer segmentation was based on what the customer would value in terms of how the intermediate storage could solve their problems, and how it could add additional value that was not problem-related. This approach was aligned with a combination of Gupta et al., (2023) definition of demographic segmentation and psychographic segmentation. Demographic segmentation focuses on who the customer is, i.e. in this case what type of business the goods recipient was running, while psychographic segmentation focuses on subjective attributes, i.e. in this case what they would value for their goods deliveries.

5.3.2 Deeper problem and concept evaluation for each goods recipient segment

After the general concept solutions were done in the second concept iteration, an evaluation was done of how well the respective concepts solve the problems for each segment of goods recipients.

The problems affecting these actors, problem 5 through 12, were listed and graded in how severe the consequence would be and how big of an economic loss the problem would generate. The grading was done by the project members based on assumptions from the interviews. A scale of 1 to 5 was used and collectively the problems was shown to have highest consequence for cafés and restaurants, due to them generally having less resources and less staff. The segment which was shown to have highest economic loss from the problems were restaurants.

The most crucial problems among all goods recipient segments were problem 5 (Damaged goods/Broken cold chain), 12 (Shortage of goods to sell), 6 (Delayed delivery to the recipient), and 9 (Unsustainable strain on staff), as they were ranked highest in severe consequences.

The concept solution evaluation was executed by ranking each concept's likeliness in preventing the problem from occurring as well as addressing it if it would occur. This was done in order to assess which of the concepts were most likely to be the suitable for each goods recipient segment. Restaurants were found to be the most crucial segment for the intermediate storage to both prevent and address the problems. B1 was the concept found to be most likely to prevent and address the problems. The specific grading in the problem and concept evaluation sheet can be viewed in Appendix F.

5.3.3 Cost and revenue analysis for the reference company and all customer segments

In the cost and revenue analysis conducted for the reference company and all customer segments, there had been identified the different types of revenue that could occur, the different types of costs, and the form of each cost, i.e. one-off costs, fixed operating costs, semi-variable operating costs, or variable costs. The form of the costs varied depending on if they belonged to the reference company or the customers, e.g. costs for purchasing a storage would be a one-off cost for the goods recipients since they would only purchase a storage once, while it would be a variable operating cost for the reference company since they would have to continuously purchase more storages following the demand on the market. Revenue was not calculated, only what financial benefits the reference company would get from selling the service and for the customers from purchasing the service in terms of where money would be made or saved.

The variables were compiled into one table for each business model, i.e. four tables were created for each of the stakeholders. Figures were collected from desktop research, interviews, and e-mail conversations. The types of costs where figures were identified were the costs for unmodified storages of different sizes, cost for parking spaces on public streets, cost of land use in public areas, and cost for parking space on real estate companies' private property. The identified figures have been omitted from this report since they summarise an investment proposal, which is sensitive information.

A large number of figures could not be found, however, and were therefore left blank in the tables. The unidentified costs for the reference company were cost for electricity, IT-system, maintenance, insurance, storage delivery, extra equipment, and storage modification. Unidentified costs for all combined customers were electricity, extra equipment, delivery costs, storage relocation, and insurance. These costs will not be as large or significant, but they will still have an economic impact, which is why they have been documented in this section so that they can be taken into account in further work.

5.3.4 Finalised technical solutions

The final concept involved the same technical solutions as in the second iteration. The idea was to have modular business models and technical solutions for a hybrid product/service offering. This final concept highlighted the importance of not having a one-size fits all service but rather flexibility as a service provider in order to meet different needs by the possible customers to the service. The concepts' technical aspects from the prior iteration would be offered in a tailored manner with a complementary enabling business model.

The technical solutions were iterated slightly from the second concept iteration with the main difference being that they would be modular storages of the same design but with different sizes for different requirements and placements for a potentially cheaper production cost. They were also renamed to make it easier to separate the solutions. In the second concept iteration, the technical solutions were divided into two main concepts, A and B, with two variants, A1 and A2, and B1 and B2, in each main concept. This was removed and instead there were four separate technical solutions with separate names. The name change was as follows: A1 became A, A2 became B, B1 became C, and B2 became D.

Technical solution A would be storages placed on municipal ground consolidating goods for multiple recipients. A sketch of technical solution A can be seen in Figure 3.

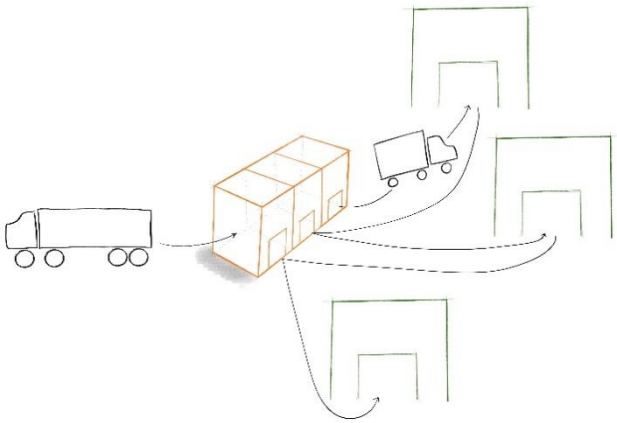


Figure 3. Sketch of technical solution A, storage on municipal ground with multiple recipients, created by the project members.

Several iterations of AI-visualisations were made in ChatGPT to provide more clear representations of the storages, including a background to provide a context for where the storages could be located. The prompts that were made to create all visualisations can be seen in Appendix G. Figures 4 and 5 show two different scenarios for technical solution A, where Figure 4 shows a Storage located at a metro station above ground, and Figure 5 shows a storage located in a parking space on a municipal street.



Figure 4. AI-visualisation of technical solution A, created in ChatGPT.



Figure 5. AI-visualisation of technical solution A, created in ChatGPT.

Technical solutions B and D would be storages placed on real estate companies' land, consolidating goods for multiple recipients, either positioned free-standing, or against a loading door. A sketch of technical solution B and D can be seen in Figure 6.

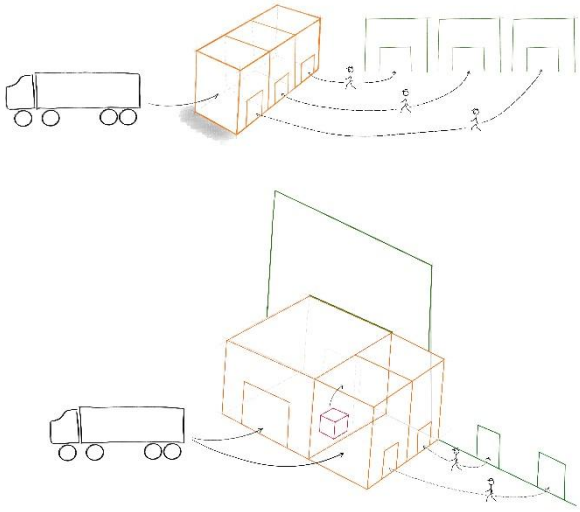


Figure 6. Sketch of technical solutions B and D, storage on real estate companies' ground with multiple recipients, created by the project members.

AI-visualisations were made, where Figure 7 show a scenario for technical solution B where the storage is located in a parking garage, and Figure 8 show a scenario for technical solution D where the storage is located at a real estate's courtyard.



Figure 7. AI-visualisation of technical solution B, created in ChatGPT.



Figure 8. AI-visualisation of technical solution D, created in ChatGPT.

Technical solution C would be storages placed on real estate companies' land, storing goods for one single recipients, positioned against a loading door. A sketch of technical solution C can be seen in Figure 9.

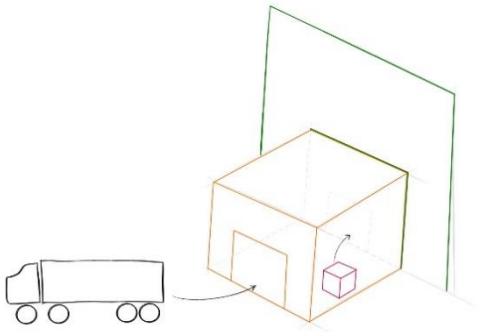


Figure 9. Sketch of technical solution C, storage on real estate company's ground with one recipient, created by the project members.

AI-visualisations were made for technical solution C, where Figure 10 and Figure 11 show two versions of a storage located at a real estate's courtyard.



Figure 10. AI-visualisation of technical solution C, created in ChatGPT.



Figure 11. AI-visualisation of technical solution C, created in ChatGPT.

5.3.5 Finalised business models

Four business models were developed which were called *model 1*, *model 2.1*, *model 2.2*, and *model 2.3*. This division was made because the main difference is that model 1 has only one customer, while model 2 has 3 customers, which is why it was divided into 2.1, 2.2, and 2.3. This section is a summary of all business models, and it structured after the nine categories included in the Business Model Canvas: *Customer segments*, *Customer relationship*, *Channels*, *Value proposition*, *Key Activities*, *Key Resources*, *Key Partners*, *Costs*, and *Revenues*. The structure follows Interaction Design Foundation's (2021) guide to how to create a Business Model Canvas. The business models are from the reference company's point of view, i.e. what

are their customer segments, what value propositions the reference company would give to their customers, what costs the reference company would have, etc.

Business model 1

The customer segments in business model 1 are logistics companies. The reference company's customer relationships centre on the possibility that they can sell this service to their current customers who purchase their trucks as an additional service or as a package deal if they buy new trucks. They can get new customers by marketing this new offering on their company website, as well as from public pilot projects where they invite several logistics companies who can try out the service. The reference company can reach customers through interactive communication channels, such as the public pilot projects with plenty of marketing. The values that the concept would offer the customer in this business model are efficient deliveries, standardised off-loading process, no boom driving if the goods recipients has the possibility of unmanned reception, better customer relations, more customers, and more orders. The key activities are managing contracts and partnerships, and maintaining infrastructure of the storages and their operation. The key resources needed are an inventory of storages, personnel performing the initial delivery of the storages, personnel performing maintenance, IT-systems. The key partners are municipality A and storage manufacturer/supplier. The cost drivers are purchasing storage units, modifying the units, placement areas, labour hours, insurance, electricity, IT-systems, maintenance, delivery of the storage. Lastly, the revenue streams are sales of the service and potentially sales of insurance if the customers choses to include it.

Business model 2.1

The customer segment in business model 2.1 are logistics companies. The reference company's customer relationships centre on the possibility that they can sell this service to their current customers who purchase their trucks as an additional service or as a package deal if they buy new trucks. They can get new customers by marketing this new offering on their company website, as well as from public pilot projects where they invite several logistics companies who can try out the service. The reference company can reach customers through interactive communication channels, such as the public pilot projects with plenty of marketing. The values that the concept would offer the customer in this business model are efficient deliveries, standardised off-loading process, no boom driving, better customer relations, more customers, and more orders. The key activities are managing contracts and partnerships, and maintaining infrastructure of the storages and their operation. The key resources needed are an inventory of storages, personnel performing the initial delivery of the storages, personnel performing maintenance, and IT-systems. The key partners are storage manufacturer/supplier and real estate companies. The cost drivers are purchasing storage units, modifying the units, placement areas, labour hours, insurance, electricity, IT-systems, maintenance, and delivery of the storage. Lastly, the revenue streams are sales of the service and potentially sales of insurance if the customers choses to include it.

Business model 2.2

The customer segment in this business model 2.2 are real estate companies. The reference company's customer relationships initiate with informing of this service to their current customers who purchase their trucks, who in turn can investigate interest among their customers. They can get new customers by marketing this new offering on their company website, as well as from public pilot projects where they invite several real estate companies

who can try out the service. The reference company can reach customers through interactive communication channels, such as the public pilot projects with plenty of marketing. Value propositions for the customers are that they get more attractive environments surrounding their facilities during the day. In addition, by investing in this service that improve their tenants' businesses', they will benefit from longer lease durations, reduced vacancy rates, and more stable rental income. The key activities are managing contracts and partnerships, and maintaining infrastructure of the storages and their operation. The key resources needed are an inventory of storages, personnel performing the initial delivery of the storages, personnel performing maintenance, and IT-systems. The key partners are storage manufacturer/supplier. The cost drivers are purchasing storage units, modifying the units, labour hours, insurance, IT-systems, maintenance, delivery of the storage, and purchasing/production of additional equipment. Lastly, the revenue streams are sales of the service and potentially sales of insurance if the customers choses to include it.

Business model 2.3

The customer segment in this business model are goods recipients, i.e. grocery stores, cafés, and restaurants. They can get new customers by marketing this new offering on their company website, as well as from public pilot projects where they invite several actors from these customer segments who can try out the service. The reference company can reach customers through interactive communication channels, such as the public pilot projects with plenty of marketing. Value propositions for the customers are more reliable deliveries, lower labour costs, less strain on staff, more sales revenue, less lost sales, and better customer relations. The key activities are managing contracts and partnerships, and maintaining infrastructure of the storages and their operation. The key resources needed are an inventory of storages, personnel performing the initial delivery of the storages, personnel performing maintenance, and IT-systems. The key partners are storage manufacturer/supplier. The cost drivers are purchasing storage units, modifying the units, labour hours, insurance, IT-systems, maintenance, delivery of the storage, and purchasing/production of additional equipment. Lastly, the revenue streams are sales of the service and potentially sales of insurance if the customers choses to include it.

Involved stakeholders and their roles in each business model

Each stakeholder was assigned to a specific role in all four business models according to the identified roles that are commonly used in logistics ecosystems, i.e. *orchestrator*, *customer*, *complementor*, *supplier*, *regulatory authority*, and *indirect stakeholder*, as described in Section 4.1.4. Table 2 presents what role each stakeholder was assigned in each business model.

Table 2. Stakeholder roles in each business model

Role	Model 1	Model 2.1	Model 2.2	Model 2.3
Orchestrator	The reference company	The reference company	The reference company	The reference company
Customer	Logistics companies	Logistics companies	Real estate companies	Cafés, Restaurants, Grocery stores
Complementor	Municipality, LMD company, Parking company	Real estate companies	Logistics companies	Logistics companies
Supplier	Goods producers	Goods producers	Goods producers	Goods producers

Regulatory authority	Municipality, Unions	Unions	Unions	Unions
Indirect stakeholder	Cafés, Restaurants, Grocery stores Society	Cafés, Restaurants, Grocery stores Society	Cafés, Restaurants, Grocery stores Society	Real estate companies Society

In general, the stakeholders are mostly changing who does what with each other and who is the customer. But one important thing to notice is that the reference company have the same role in every model, the orchestrator, which is usually the leading role in an ecosystem, and they will be the ones selling the service in every model.

Modular product/service-hybrid offering

The end result of the third concept iteration was a modular product/service-hybrid offering where the four different business models could be combined in six different combinations with the four different technical solutions. Technically, more combinations are possible, but some were excluded because they were not considered at all feasible or viable, based on the business knowledge and logical thinking of the project members. Therefore, the decision to only include the combinations that were considered more reasonable was made to remove ideas that would never be realised anyway. The combinations are presented in Figure 12.

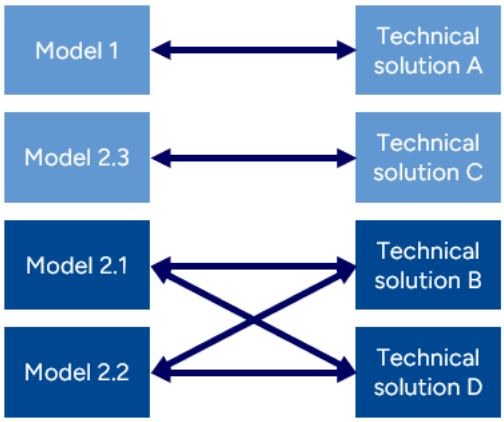


Figure 12. Combinations between business models and technical solutions.

The reason for creating a modular business offering was to reach a larger market with several customer segments for the reference company where they could tailor the concept to their customers' specific needs, requirements, and conditions.

5.4 Observations in the local environment

This section presents the field observations that were made in different districts of central Stockholm. The reasoning for the selected districts are presented, as well as what observations that were made in respective district in terms of geographical possibilities for placing a storage, and what problems and needs grocery stores, cafés, and restaurants may have related to goods deliveries caused by traffic and street conditions. The observations were made to analyse the practical potential for the developed concepts in the second iteration.

5.4.1 Observed urban districts in central Stockholm

There were four urban districts in Stockholm that were observed: *Gamla stan*, *NoHa*, *Östermalm*, and *Medborgarplatsen*. These districts were chosen due to them having both differences and similarities. The main differences that were aspects relevant to be observed were due to their diversity in environmental setting, different architecture, and different infrastructure. They also have some relevant similarities, such as that all of them are very close to the city centre, and due to that all of them are highly populated with stores and restaurants, which could be potential future users of an intermediate storage. The scope of the project was limited to central Stockholm, which is why only districts that are close to each other were selected instead of some further away. Since an overall aim of the project was to gain as much insights and perspectives as possible from many different angles, these districts were determined to be suitable to investigate due to them having both similarities and differences for comparison purposes.

5.4.2 Intermediate storage geographical possibilities in respective districts

Despite each of these districts being located relatively close together, they have some very distinct differences and possibilities to physically incorporate intermediate storages.

Gamla stan is a small island located just south of the city centre of Stockholm. It has very old architecture and infrastructure with very little surrounding space. Due to the way the district is built, there seemed to be no possibilities to place storages inside Gamla stan, i.e. technical solutions B, C and D would not be possible to implement there. However, on the outskirts of the island there are several areas where storages could geographically fit in such as alleys and parking spaces, and as the island is relatively small, it should be sufficient to place storages only on the outskirts to enable technical solution A.

Norr om Hamngatan (NoHa) is a small area that is a part of the larger district Norrmalm, which is the most central area in Stockholm. It is a very densely populated area with a varied layout of streets. Because of this, there are not many opportunities to place storages that can be shielded from the public, which is necessary for technical solutions B, C, and D because the goods recipients have to go out and collect their goods from the storages themselves and therefore need to be protected against theft. However, there are plenty of parking garages where there could potentially be opportunities to place storages shielded from the public, but this was not investigated further and could therefore not be confirmed. There are some alleys and empty spaces between buildings where there is a possibility to place storages that can be used for technical solution A.

Östermalm is one of the most luxurious districts in Stockholm, east of the city centre, where there is a mix of roads with densely built houses and more open spaces with parks and public squares. The observations were mainly carried out around Östermalmstorg which is a very central part of Östermalm. There did not seem to be much unutilised space where storage could be placed at open areas, except perhaps at parking spaces along the streets which are, however, not very wide and it would therefore be not a lot of room for a storage i.e. it would be difficult to implement technical solution A in that area. There also did not seem to be many courtyards where trucks could enter, which means that it may also be difficult to implement technical solutions B, C, and D.

Medborgarplatsen is an area located in the larger district Södermalm in the southern part of Stockholm. The architecture at and around Medborgarplatsen has many large blocks of buildings, several of which has secure courtyards within them where there is the possibility to drive in a truck. This means that it should be possible to place storages in these areas that are inaccessible to the public, where goods can be moved safely. This means that technical solution B, C, and D should be possible to implement in that area. Since the streets are very wide with plenty of parking spaces, there should also be possible to place storages there, i.e. technical solution A should be possible to implement.

5.4.3 Problems and needs in the grocery store, cafés, and restaurants market segments in terms of goods deliveries in respective district

Due to each district having several differences in terms of architecture, infrastructure, business density, and population, the problems and needs also differed.

Gamla stan is the area that were identified to have the largest problems in terms of goods deliveries. The streets are very narrow and there is an overall problem with insufficient space or lack of proper loading spaces for trucks to perform deliveries. There is also the issue with that the streets are very crowded with residents and tourists since it is a popular tourist area. Access restrictions for drivers performing deliveries due these factors is a commonly occurring problem, as can be seen in Figure 13 where there is access restrictions on the street due to the amount of people, and in Figure 14 where a delivery is being performed at a hazardous way due to the lack of a proper loading space, which in turn made the truck block essentially the entire street.



Figure 13. Access restrictions for a logistics company due to crowded street in Gamla stan. Picture taken by the project members.



Figure 14. Suboptimal loading of goods in Gamla stan where the truck was blocking the street. Picture taken by the project members.

The reason for looking into NoHa specifically and not the entirety of Norrmalm was due to the pending integration of environmental zone 3, which was supposed to be taken into effect in 2024 but is currently temporarily on hold (Stockholm Stad, 2024). Either way, why that was interesting to consider was due to the transportation restrictions that will be implemented if the environmental zone will take into effect. Regular diesel trucks will no longer be allowed there at any point of the day, which will affect how goods deliveries will be made. Stores and restaurants may be affected by this if their normal suppliers do not have electric trucks or other authorised vehicles, which may lead to them having to switch suppliers, which in turn may lead to negative economic consequences. Another aspect of road conditions in NoHa is that there are many large streets with a lot of traffic where the road layout is not optimal in some places for a smooth traffic flow. One of the observations in this area was that the traffic was very heavy despite that it was around 14:00 o'clock at the time, which is not a typical high peak hour. According to one of the interviewees in this area, they usually received their goods at around this time, which can cause large problems with delayed deliveries. The need for off-peak deliveries in this area was discerned from these observations as high.

At Östermalm, and particularly around Östermalmstorg, there are many smaller shopping streets with a mix of stores and restaurants where a lot of people are moving around. Because of this, it can be difficult for trucks to get around, which can lead to delayed deliveries causing problems for goods recipients. However, the number of cars and other vehicles not making heavy deliveries is much smaller compared to NoHa, which is otherwise relatively similar in terms of type of architecture and road infrastructure. Therefore, the need for more storage space is probably not as great in Östermalm as in NoHa, as the main problem is pedestrians who do not take up as much space as cars.

Medborgarplatsen were observed to have a high density of all three market segments, as well as traffic and pedestrians at all hours of the day. However, this area is generally relatively well

planned, with open areas and loading zones which allows deliveries to be made on time. Therefore, the need for storages in this area is probably the smallest of all observed districts.

5.5 Assumptions

There were a couple of assumptions made during the project. First, the cost of storages. The assumption was based on sources available on the internet by retailers' information regarding their price for either purchasing or renting containers of various types and sizes. The assumption was that the cost of the storages in the concepts would be relatively similar to the current market prices. Another assumption related to the storages was what size would be sufficient for efficient use, i.e. it should be able to move around and fill goods for one or more recipients. A third storage related assumption is the cost of potential placement areas for the storages. They were based on data retrieved from interviews, e-mail conversations, and desktop research, and the assumption was that the cost for placing the storages would be relatively similar to those figures.

Another assumption made was the grading of the problem analysis for the recipients and how they would evaluate the problems. This assumption was based on qualified guesses by the project members.

Finally, the concepts require a number of criteria to be realised regarding, for example, permission to place warehouses, permission to carry out night deliveries, noise level, and possibility of power supply. The assumption that was made was that these factors would be solvable and not be an obstacle. Of course, all of these factors are critical to take into account in further work, but they were low priority to take into account during this project.

5.5 Findings from concept interviews

In this section the insights on concept interviews are presented. They resulted in informational findings in all of the below listed areas, by the organisational categories *grocery stores, cafés, restaurants, logistics companies*, and *real estate companies*, and in total, 16 concept interviews were conducted.

5.5.1 Different perspectives on deliveries

Delayed deliveries were not a typical problem according to some respondents from grocery stores, cafés, and restaurants. It could happen, but it was never mentioned as a meaningful issue. In general, the stores and restaurants that were interviewed claimed that deliveries mostly were on time.

Many respondents were positive towards receiving goods before opening and being able to handle it before their customers arrive, as they would get a more effective workload. Restaurant C receives goods before they open with unmanned reception, which they think is very beneficial as they often were able to process a delivery before customers arrive, but on occasion, they would have to make the customer wait while the delivery was handled if they did not have the time to finish it before. Grocery store A unpacks goods during the day and has a lot of staff on site at the same time, so it is not a problem to handle goods while customers are there. However, the respondent was positive about unmanned delivery before opening if it could be made possible.

There were also respondents that were hesitant towards unmanned reception. The respondent from café C claimed that they have to be present when deliveries arrive and do not have the possibility for unmanned reception. The respondent from restaurant B also said that they have to be on site to receive deliveries and cannot give out a key to the supplier. The respondent from restaurant A added that they think it would be difficult with unmanned reception at night when they deliver fresh and frozen goods because that must be received and unpacked into the refrigerator and freezer directly.

From the real estate companies' perspective regarding deliveries, real estate company A mentioned the issue of manned receiving of goods if the deliveries were to be done during off-peak hours. Especially on properties with no loading area. Real estate company B would have difficulty enabling logistics companies in general to deliver at night. Today they only use one operator, and if they need to enable several companies, they must have some new type of passage. It was perceived as difficult both practically and financially.

5.5.2 Opportunities and possibilities in the concepts

Grocery store A were very positive towards getting more storage space and unmanned delivery before staff was on site. They were also positive to increasing retail space. The respondent from café A agreed that it would have been nice if the deliveries came before the morning rush hour, but they do not see a greater value in getting deliveries earlier than they get today as customers do not come until several hours after they open. However, the respondent believes that there is a value in general for cafés to get deliveries earlier in the morning before rush hour. The respondent from café C saw an opportunity with the concepts in that staff could arrive at the store before they open and would therefore have time to take care of the deliveries. They could adjust so that the staff would be on site earlier according to when the delivery would arrive.

From the respondent from logistics company D, there was a positive attitude towards the concepts in general. There was an expressed positivity towards off-peak hour work, only performing LMD, or owning the intermediate storage. Logistics company E expressed an interest in concept A1 where they could deliver the goods from the intermediate storages to the recipients. They were very positive toward off-peak deliveries from their point of view but highlighted that the goods recipients need to adapt then. There was a slight wonder from logistics company C with refrigerated goods delivery as it is not done by them today. An interest of intermediate storage ownership and maintenance was expressed by them. A requirement to ease adoption of this specific actor was to include returns in the logistics chain of the concepts.

The respondents from real estate company A expressed that they value a good environment for their tenants, thereby having an interest in reducing heavy traffic daytime.

> *"And our properties and those who sit in our properties, they feel good if we create street spaces and places where people want to be and to create like street spaces and places where people want to be and part of that is then to like get the traffic off the street simply in one way or another either get, like yes, but replace heavy vehicles that make a lot of noise with lighter vehicles that make less noise and take up less space in the street environment or be able to get deliveries at times when like those who move in our places and our houses do not move in*

> *these places."* (Translated by the authors, anonymous participant, personal communication, April 29, 2025)

They also mentioned an opportunity in the concepts in that placing an intermediate storage on their properties, if space allows for it, was possible. They recognise more values associated with removing heavy traffic during peak hours for their tenants making them less focused on having to make money on the business.

> *"That it could be done at times when it does not compete with with like something else would be great. But both for for example those who choose to drive to work, that there will be like less less traffic, but also, yes, what I was talking about earlier, like the general experience in the street environment."* (Translated by the authors, anonymous participant, personal communication, April 29, 2025)

The most promising business model for real estate companies were expressed by real estate company A to be them renting out space to put intermediate storages on as this would not create new business activities for them.

> *"We don't want to be a service provider in the sense of like selling a service. The simple answer is then probably that it is easiest for us to rent out."* (Translated by the authors, anonymous participant, personal communication, April 29, 2025)

Real estate company B has many properties without courtyards or other accessible loading bays. In the few properties that already have loading bays, receiving goods is not as much of a problem, not for them in relation to their customers.

> *"The disadvantage then is that these are of course, these are places where we have loading bays as it is with goods receptions. In principle, all places so it's like the same places. So, there it is a, a solution for goods deliveries actually already in these places."* (Translated by the authors, anonymous participant, personal communication, May 2, 2025)

Real estate company B's perspective on the business model for their part was that they see an opportunity in buying the service from the service provider, which they in turn would sell to their tenants. The advantage of having the service is for them to be able to show and sell the complete solution to their customers by including it in the rental agreement. They also validate and share the perspective real estate company A had that the advantage of renting out land is that it is worry-free for them and a guaranteed income. But they mention that the disadvantage is that it may be more difficult to sell the concepts to their tenants.

> *"For the sake of simplicity, purely like the practical work every day, I would say it would be so much easier for us to just rent the space to somebody and say, fix it. You pay us rent and you deliver the service to our tenants, great. But I don't think we will get anywhere with that, unfortunately."* (Translated by the authors, anonymous participant, personal communication, May 2, 2025)

6. Results from validation phase

This chapter presents all the results from the validation phase, including findings from the validation interviews which provides comments regarding the sustainability, feasibility, profitability, and viability of the concepts.

6.1 Findings from validation interviews

The validation interviews resulted in informational findings in all of the below listed areas, by the organisational categories *truck manufacturer*, *logistics companies*, *governmental authorities*, and *unions*, and in total, four validation interviews were conducted.

6.1.1 Problems goods recipients currently experience

The presented challenges, of delayed delivery and strain on staff connected to deliveries was confirmed by the respondent from grocery store B to be occurring issues. Delayed delivery had been an issue where some orders could be up to a day late. The respondent emphasised that it is crucial for them to ensure that their customers have the best possible experience of their store, and an important aspect of that is to always have products for sale. Delayed deliveries can sometimes make them loose customers, but it was difficult for the respondent to quantify the amount of loss due to delays. Broken cold chain of goods during transport was not a problem they had experienced. However, damaged goods have happened where the driver does not notify about it, which makes for a burdensome process of receiving compensation, whereby they only get compensated for the price of the goods and not for the inconvenience or the lost sales. Similarly, grocery store C confirmed that the challenges were present for them as well and estimated that delayed deliveries cost them about 1-2% of their annual revenue. The respondent from restaurant D also confirmed that the challenges of delayed delivery, strain on staff and broken cold chain affected them. Occasionally they have received goods that were not as cold as they were supposed to be, but it was not a recurring problem connected to any notable costs for them. It was mentioned by the respondent that it would be great to get deliveries earlier on the day for them to be able to offer their customers the dishes on the menu as planned.

6.1.2 Challenges with the concepts

Logistics company D expressed challenges with getting goods recipients to open their businesses earlier if manned reception is required by the concepts. Also, that the logistics company have to guarantee goods deliveries by a certain time in the intermediate storage to avoid delayed reception of goods. The increased cost of transhipment and guaranteeing shortened lead time were expressed as potential hinders for the concept.

One challenge expressed by the respondent from the Swedish Police Authority connected to the concepts were the potential for stealing the entire storage for the concept of placement on the municipality's ground. Technical solutions, like RFID or temporary locking mechanisms, would have to be utilised in order to mitigate this risk. However, the risk of these storages being stolen were considered low as the goods aimed to be stored in the storages was not considered a goods type subject to theft. Two solutions were mentioned to mitigate the risk of thieves being interested in the storages, either by anonymising the deliveries completely, which makes it impossible to know what specific goods are being stored, or the opposite by providing

information of the type of goods being stored in the storages, which could make potential thieves disinterested in them.

In the interview with the Swedish Transport Workers' Union, it was highlighted by the respondent that the aspect of off-peak deliveries in the concepts could meet resistance from their members and them as a union as night work has shown to shrink life length expectancy and is considered a bad working environment. However, the stress factor of today's deliveries, working shifts being prolonged due to e.g. access restrictions and delayed deliveries, was highlighted as a commonly mentioned issue impacting the workers well-being and would be beneficial to overcome. They would not accept placement of the storages in vulnerable areas with the reflection of the workers carrying out the goods transshipments alone and therefore in an exposed position.

A presentation of concept A and business model 1 was made for municipality A, where the participants saw possible challenges with how the presented concept would be financed. The loss of confirmation from logistics companies to be a customer and pay for this service made the participants unsure of the feasibility of the concept economically. They also mentioned that a challenge with the concept would be to identify underutilised placement areas other than parking spaces, due to lack of physical space as well as regulations. Another challenge was depending on how many storages there would be, there could be a potential issue identifying and reserving enough parking spaces.

6.1.3 Adoptability of the concepts

According to the respondent from the Swedish Police Authority, there was most likely a low risk associated with increased crime rates or decreased security for involved actors by the introduction of the concepts, despite the off-peak aspect. However, in the interview with the Swedish Transport Workers' Union, the respondent mentioned that in order for the off-peak delivery aspect of the concepts to be regarded at all was if the workers were compensated in some way, either by monetary compensation or by offering shorter working hours per shift.

The respondent from grocery store B was positive towards technical solution A and was open to rescheduling their staff earlier if the values of the concept was ensured. They were also positive towards the idea of enabling for a logistics company to have access and placing the goods in their storage spaces during the night. As this grocery store had several suppliers, they receive a lot of small deliveries and the potential of this concept to reduce the amount of deliveries by increasing the volume of goods in one delivery was highlighted to be interesting and valuable.

The respondent from restaurant D was somewhat positive towards the concept. For them to adopt the concept there would have to be an infrastructure where all actors benefited. They were interested in the concept and adapting to it if the concept would bring other values on a societal level. However, some reservations towards the concepts were expressed as fearfulness of longer delivery durations, more incorrect handling of the goods, more traffic during night, and working conditions for the transporters working off-peak hours. If there would be an incentive for the restaurant to order goods with better foresight, where they could buy their goods at lower prices earlier in the day, the respondent believed they would be more encouraged to receive the benefits and values of the concepts. The respondent from grocery

store C also expressed a reluctancy towards the off-peak aspects of the concepts. Concerns about residents in the facility were expressed, with regards to them being disturbed by deliveries and handling of goods at night-time. Another challenge from their perspective was to reschedule staff earlier in the day as their contract with the real estate company states that they cannot handle goods and receive deliveries earlier than 7:30, due to complaints by residents in the same building. Unmanned reception was deemed impossible by the respondent as their storage space was very limited making them in need of handling deliveries when they arrive.

In the presentation for municipality A, the participants validated that the concept of rotating the storages was a good idea, and that it could be a loophole to make the provision of the placement areas easier. The participants believed that parking spaces would be the most feasible placement areas for storages as other underutilised areas had been hard to identify in prior similar projects. They mentioned that there would be highest potential in starting in a small scale and only focus on Gamla stan. Apart from parking spaces, it was expressed that there would be an easier direction to utilise privately owned areas such as real estate areas instead of municipal areas, as suggested by the other concepts. One participant from municipality A made it clear that they would be interested in participating in a concept similar to the one presented with an external orchestrating actor. It was made clear that municipality A would not be interested in financial compensation for resources to execute their part in the business model, although placement areas in the form of parking spaces would have to be financed by the orchestrator. It was mentioned that for the concept to be interesting for municipality A to participate in, there would have to be possible for multiple logistics companies and goods recipients to be customers to avoid monopolisation.

One respondent from the reference company's venture business department validated the feasibility of business model 1, which is the most complex, ensuring that this could be an interesting business to invest in. The concepts were said to possibly be complemented to align with their aim to make sustainable developments in transportation. It could elevate the marketability and attractiveness of the concepts for all parties, especially the reference company, municipality A, and logistics companies. The respondent also validated the necessity to provide modular technical solutions as well as modular business models in order to have flexibility in the service provided such that more customers can be reached, and market changes can be adjusted for.

6.2 Key problem validation

During the validation interviews with the three respondents from different goods recipients' market segments, quantifiable data on how the key problems identified were experienced by them was acquired. Figure 15 illustrates how many of the respondents confirmed each problem.

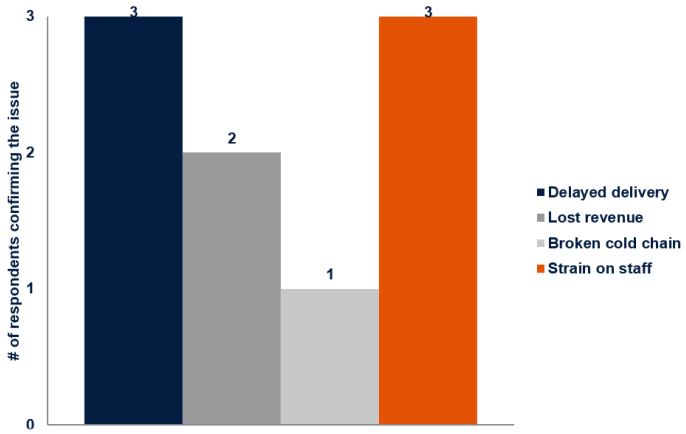


Figure 15. Statistical presentation of the validation results of key problems by the respondents from goods recipients.

The problems were validated by the respondents and both delayed deliveries and strain on staff were problems confirmed by every respondent, further highlighting the importance of addressing those problems and improving those areas with the new concepts.

7. Analysis and Discussion

This chapter presents an analysis and discussion of the project results, the existing knowledge, the validity and reliability, and recommendations for future research. Whether the research questions have been answered have also been analysed in this chapter.

7.1 Analysis of the project results

This section presents an analysis of the innovation process, some important practical results from the project, and discussions regarding what contributions the project results have on existing research related to the project. In addition, there is an analysis of what the project results have on the economic, social, and environmental aspects to society and industries.

7.1.1 Analysis of the innovation process

The innovation process of the project was initially solely based on using the Design Thinking process model. However, in an early stage of the project, it was realised that due to the large scope and the nature of the project, Design Thinking was not sufficient as guideline throughout the project. Therefore, a decision was made to establish and include three new sub-phases, research phase, innovation phase, and validation phase, under the Design Thinking phases as described in Section 3.1.1. and Section 3.2.1. The nature of the project required several different activities being carried out in parallel, which resulted in the need for an approach with several short iterations. The Design thinking model is structured with more defined phases than what was desired, which made it difficult to place the activities in a clear order. Therefore, the three sub-phases were created to allow a more flowing process with less defined activities. The different types of interviews were a particular reason for the need for different set of phases. Interviews were made throughout the entire course of the project and they were all conducted for different reasons. It did not fit to place interviews in only the emphasise phase, which was originally the plan, since the interviews had more purposes than just understanding the users. This made it make sense to differentiate the phases accordingly.

The new structure that was defined for the project phases had a very successful outcome. In particular, it became much easier to distinguish between the different types of interviews and their different purposes, as well as the value that the interview results contributed to the project. It also allowed for less restrictions related to

In retrospect, the Design thinking model was not particularly useful for this project. The new structure was much more useful according to the project members. However, the idea for that structure was based on Design thinking and that it did not meet the needs for the project according to the project members, which made its contribution valuable. Based on this insight, it was concluded that it can be useful to redefine existing process models and tailor them to better fit the nature of a project, as well as to take into account personal preferences regarding ways of working.

The Sprint model served a purpose of regularly providing a chance for the project members to look back during the semi-weekly sprint meetings at the progress and assess both what obstacles the project was facing, but also what had been successful. It was a tool to assess what pace the project was moving at, however, the usefulness of the Sprint model was only somewhat necessary during the course of the project. It was especially useful in the early stage

of the project when the scope and tasks were a bit more diffuse. Later in the project, assessments of project progress were made more continuously through discussions, making sprint meetings and documentation less necessary.

One consequence of using the Sprint model was that sometimes it was overestimated how much could be done in the two-week sprint period. Initially, planning in two-week intervals seemed like a reasonable time span, but it quickly turned out that it was not such a good structure as the project was so iterative and sensitive to changes that happened in the planning all the time. Therefore, it was not only a problem of estimating the number of tasks for a sprint, but also that if a certain task could not be completed due to some external factor, other tasks that depended on it also could not be performed. Therefore, it was decided that plans with shorter intervals were needed, but also plans with more concrete tasks. As a result, a joint to-do list was created in Microsoft Teams, as well as an individual one for each project member which were updated almost daily. This helped with planning as it allowed for more iterative work but with clearer tasks. It also provided confirmation of progress over time when completed tasks could be ticked off, which also brought less negative feelings when the two-week plans could not be completed. In summary, the Sprint model did not fit the nature of the project or the preferred way of working of the project members.

7.1.2 Analysis of the practical project results

This section presents an analysis of the practical project results in relation to desktop research and possibility for adoption.

Analyses between developed concepts and desktop research

The need for fast delivery times and personal deliveries of today expressed by Giuliano (2023), indicated that there could be a reluctancy to consolidate goods, if that would mean longer delivery times and a less satisfied end customer, impacting all actors in the system. There was therefore thought to be highly important to ensure technically that goods are not delivered more slowly due to the consolidation and off-peak aspects of the concepts. An alternative solution could be to communicate to the goods recipients the additional values and benefits this new system would bring not only to them but for the other actors which in turn possibly would create more flexibility from them with regards to delivery times.

The success of the existing B2C solutions presented in Section 4.2 that were similar to the hypothesised solution of this project, suggests that there are possibilities to make improvements to the logistics chain between businesses, with the main value of reducing cost of transportation and increase the sustainability aspect of the deliveries by consolidating goods and reducing transport distances. The concept was also considered to be interesting to investigate due to the emerging distribution network of smaller hubs closer to the cities. Many logistics companies and other businesses want to have their goods closer to their customers in order to reduce delivery times which indicates an emerging willingness to have storage facilities closer to the goods recipients.

Storage of refrigerated goods was discovered to be challenging and limited in the ways of achieving it. Perishable goods were identified in the desktop research by, among others, Shashi et al. (2020) as a category of goods that has many rules and requirements regarding storage and transport. This makes it more crucial to have technical solutions guaranteeing a maintained

cold chain. The issues with packaging logistics and refrigerated goods transport was confirmed by the respondent from university A lead the project members to believe that a new type of packaging had to be developed and used when transporting refrigerated goods in order for it to be stored in unrefrigerated storages, like for example smaller thermoelectric cool boxes. However, in the end, no decision was taken on the technical solution to be used to keep the goods refrigerated, as not enough data was produced on what would be most appropriate.

Analyses of the potential for adoption of the concepts

With regards to adoption of the system innovation, the interview with the respondent from research centre A highlighted that main focus should be on early adopters while the reference company mentioned that focus should be on late majority when innovating an entire system. This interesting contradiction in the results indicated that early adopters are important to focus on when introducing new technology in order to get early traction in the market. The late majority, on the other hand, was thought to be important to consider when building new innovations in a system in order to ensure long-term success of the innovation. As this project is about a new system change more than the introduction of new technologies, the late majority should be the focus. These actors are likely to be goods recipients who would potentially be reluctant to rethink their process for receiving goods, as many do not see sufficient value in making that change. Communicating the value of the concepts to these specific businesses will therefore be crucial for the innovation to reach the scale required to ensure economic impact and sustainability. The innovation's value proposition should therefore be focused on addressing each individual's specific needs. Addressing needs were thought to facilitate adoption and it was concluded that dialogue must be held with all affected stakeholders to identify needs and communicate how the system innovation would meet those needs.

Long-term placement of intermediate storages in public areas was identified to be undesirable by municipality A whereby the technical solution of having storages being positioned on municipality's ground must involve a continuous rotation of the storage in order for them not to take up one given spot permanently. This was considered when developing the concepts and is a recommended factor to consider in other variants of a technical solution involving placement of storages on municipalities' land.

Staff at interviewed grocery stores, cafés, and restaurants were overall very positive about the solutions the concepts could potentially provide. Early, punctual, deliveries are preferred, and the possibility of expanding the store area was also met with positivity. However, problems arose when the concepts were presented. The values they were supposed to convey could not be presented clearly enough, mostly because the interviews were very short. This meant that positivity towards the concepts themselves could not be validated.

The interviews with real estate companies gave contradictory results, making their position in the final business model of the system flexible, whether they could rent out space or buy the service and sell it to their tenants. It would most likely depend on which specific real estate company that would be involved as they have different views in how they want to participate in these concepts. The real estate companies that were interviews were interested in the benefits of the concepts and saw it as valuable to the attractiveness of their properties in different ways.

The Swedish Transport Workers' Union is one of the more challenging actors to satisfy in these concepts as they expressed a strong opposition towards their members working off-peak hours. Their view contradicted that of municipality A and research centre B who mentioned that people working night shifts with delivering goods usually did not want to go back to work during the day. That was an interesting contradiction, and it should be investigated further to clarify how extensive the interest to work during the night is.

The importance of clear roles in new ecosystems was identified both in the literature review by Talamar et al. (2020), among others, and in an interview with a respondent from the reference company. This was something that was taken into account when the business models were created, as roles were assigned with specific tasks for each actor involved in the system. And should this be taken into account even in further work on the project, this would likely increase the chances of adoption to occur.

7.1.5 Analysis of what the project results contribute with to the academic research on system innovation

An analysis was made in relation to Flynn's (2007) claim described in the literature review about the complexity of assessing public acceptance of new technologies, as the author argues that it can be difficult for citizens to express their opinions and give their approval before the technology is introduced. Although this project did not focus on new technologies, it can be argued that the same issue applies to the introduction of major system changes. It was difficult for some stakeholders to give an opinion on the concepts and ideas in general, as it was difficult to get them to understand the concepts when they are so underdeveloped.

The process models initially intended to be used in the project were modified to better fit the project, leading to major improvements in the work. The usefulness of this was not initially analysed, but the result was clear evidence that this should be taken into account during innovation projects. When starting an innovation project and considering which innovation management methods to use in the concept design, it may be useful to consider from the start whether standardised models are actually fully suitable or whether modifications should be made directly. This can apply to all types of innovation projects, not just those in systems innovation.

This project has shown that customer segmentation was a necessary choice in order to better analyse the market. Separating a large market segment into several smaller ones was crucial for identifying specific needs, problems, values, and conditions. One conclusion drawn from this was that it is probably important to take this into account in most system innovation projects and not to make the customer segments too broad. As shown in the literature review by Dunphy and Herbig (1995), one of the most important factors to achieve stakeholder acceptance is to be able to identify needs, and if customer segments are too broad, this becomes much more difficult. The reason for this is that it is not possible to ignore too many differences between actors as some differences are too big. As an example from the results of this project: grocery stores will probably be the only actor in the initial goods recipient segment that can cut down on their staff during the day with the introduction of off-peak deliveries as they have significantly more employees working at the same time than cafés in particular have. Had the initial customer segment of goods recipients not been separated, the value proposition would either have been incorrect and unrealisable by a very large proportion of the segment, or it

would have had to be removed altogether, which would have reduced the attractiveness of the solution.

7.1.6 Analyses of the economic, social, and environmental impact of project results on society and industry

This section presents a discussion of what different practical implications the project methods and results could have on society and on companies in industries working with system innovation in terms of economic, social, and environmental aspects.

Economic: New system innovation enabling business and market growth

It has been shown in the desktop research that similar solutions for intermediate storage exist and that pilot projects for off-peak deliveries have been carried out. However, solutions that include all aspects that have been investigated in this project, i.e. intermediate storage of smaller sizes, off-peak deliveries, and refrigerated goods, are not yet on the market. This means that this project and this system innovation could be the next step in the development of sustainable logistics solutions, which in turn could provide a new type of business with a new market for those who choose to take this further.

Social: Facilitate public adoption of system innovations

This project has identified that people do not always know what is best for them, or that they do not think certain problems are serious enough to make change worthwhile. This conclusion could be drawn both from the statements of the respondent from research centre A and the respondent from the Swedish transport administration. It is not always possible to simply tell people "*This is good for you, you need this*," since people usually need to realise the facts on their own to be able to accept that they need a change. After interviewing the respondent from the Swedish Transport Administration who described an example of creating acceptance for the introduction of a median barrier on roads to create higher road safety, it could be concluded that sometimes it takes extreme cases for people to realise that a change is required. Of course, it does not have to be as extreme as death, but a common problem among a specific group of people that they don't know how to solve could provide an opportunity for organisations to achieve adoption for a new innovation among these people. One approach to doing this could be to first conduct a needs analysis to identify a demographic *crisis area*, both in terms of a market segment and a geographical area, where the need for change is greatest. After such an area has been established, the organisation can introduce their proposed solution on a small scale, evaluate the response of the individuals concerned, and thus expand once adoption has been achieved. This is an approach that could potentially work for most types of industries, but further research is needed to confirm this.

Environmental: Facilitating off-peak deliveries in urban environments to support sustainable transport development

As has been described in this report, there are major environmental problems with today's logistics solutions for goods transport, especially linked to traffic congestion. The fact that off-peak deliveries could be a solution has already been identified before this project, but what this project has contributed with is new concrete ways to test this so that it can be scaled up in the future. Which in turn will improve the environmental problems.

7.2 Discussion of whether the results answered the research questions

This section is an analysis and discussion of the project results connected to each research question and whether they were answered.

7.2.1 RQ1: How desirable, viable, feasible, and sustainable is temporary storage of goods for the reference company to develop and implement?

From the research and project findings, the hypothesis was thought to be desirable by most stakeholders. However, some find the values such as getting deliveries before opening, desirable but not the solution as they were not able to see the future perspective where this would be possible, but rather determined that it would not be possible due to the current restrictions they have.

The most viable implementation of the technical solutions and the business models for the reference company would be to offer a modular product/service-hybrid system including all models at once. The reason for this is due to the complexity of the large system, this way they can reach a broader market and meet the unique needs by individual customers in different customer segments. They would have an orchestrating role in the system, which would be a new type of business for them, providing the possibility to get competitive advantage.

The results of the project show that that temporary storage of goods is likely to be a feasible business in the future. Many individuals and organisations want to get the benefits that these concepts can bring, and the technical solutions do not require a lot of complex technology, which is positive for feasibility. However, due to the large number of actors involved and the fact that they are all slightly different on the adoption curve, as well as restrictions that exist regarding mainly off-peak deliveries, it is difficult to determine at this stage when it would be feasible. Some restrictions that exist today need to be changed, such as permitted delivery hours, for the business to be realised. Such changes can take time, which means that it is likely that it will also take time for the business to become feasible.

Due to the relative ease of implementation for the reference company and the high scalability of these business models, they are believed to be a very sustainable business for the reference company. It does not require a lot of new or advanced technology or other large transitions in their operation to start up this business, and, they do not have to be limited to Stockholm, but can broaden their market to new areas. The modular hybrid model gives them a broad market from the start, making it easier to get the business off the ground. This business is also in line with the growing awareness and willingness to find solutions to the major environmental problems of cities with too much traffic, which is good for the sustainability of the business over time.

7.2.2 RQ2: What are the critical factors and stakeholders that affect the proposed solution?

There are several critical factors affecting the proposed final concepts which needs to be addressed. First, the characteristics, conditions, problems, and individual needs of the goods recipients vary greatly, as it largely depends on the delivery rate, the type of ordered goods, property conditions, and other constraints to which they have to adapt. There needs to be a clear dialogue during the selling process analysing these factors to determine the best solution for them.

Another important factor is that due to the system-level changes involved in these concepts, the market must be mature, and enough stakeholders must be in support of the new changes for the concepts to be adopted. As all stakeholders are affected by each other in different ways, everyone needs to be willing to embrace the changes in order for them to be implemented at all. However, the degree of how important this would be for each actor varies between the different concepts. For example, in business model 2.2 where the real estate companies are the customer, in order for them to want to buy the service, there needs to be an interest on the part of their tenants, i.e. the goods recipients, to buy the service from them and thus change their delivery routines. Whereas in model 1, if the goods recipients already have the option of unmanned reception, it will not be as big a change for them to adapt to, which makes it easier to introduce that model in such conditions.

Another critical factor is to allow for flexibility of the customer's commitment to their investment in the service, meaning that they have the option to withdraw from the service if it does not fulfil or meet their needs. This could be facilitated with having a leasing pricing model rather than one time purchase. This would provide a sense of security for the customers since they would not put as high risk in making a large investment before knowing how well it would succeed.

As identified in the literature study, one very important factor to address in ecosystems with several involved actors, especially emerging ecosystems, clear roles for the actors need to be established to ensure success. The actor responsible for organising this is the orchestrator, who has a role that can be considered as a leader in an ecosystem. As there were several complementors and indirect stakeholders involved in the ecosystem in this project, all of them need to have a clear role where they can have their needs and economic interests met in order for the concept to be established. This is where the reference company's role would become so important since they would be the orchestrator in these concepts. It would be their assignment to communicating value propositions, establish relationships, manage contracting, and establish key partnerships, which in turn places all stakeholders in their respective role. Successfully performing these tasks creates good conditions for the survival and expansion of the ecosystem. Technically, however, it would not be a completely new type of emerging ecosystem, as logistics systems with these actors already exist. However, as major changes to the previous logistics structure would be implemented with these concepts, their role becomes equally important.

It has been identified that goods recipients are a difficult stakeholder to create adoption of the type of system change investigated in this project. They have adapted their routines to the conditions and restrictions they have today, and many have become comfortable with it, which means there is a certain unwillingness to change. In other words, they are far behind in the adoption curve as a late majority, which means that it takes a lot of dialogue to get them involved in system innovations. Having a clear dialogue about what changes the innovation would bring, both positive and negative, especially being prepared to respond to their concerns could help create adoption.

7.2.3 RQ3: How can adoption among stakeholders of system innovations be facilitated?

The project has shown that strong relationships, communication, a common goal, ensured value offerings for all actors, and pilot tests proving value in the system innovation are all factors

contributing to adoption. Each individual stakeholders' specific needs, values, and conditions must be thoroughly analysed so that the best possible solution could be presented to them where those factors would be met. Iterative communication and practical experimentation of the innovation has to be performed in order to facilitate equal willingness to adopt the innovation with all actors. This will build trust and address concerns which some stakeholders might have, making the adoption easier.

7.3 Validity and reliability

There were a couple of factors in the project which affected the validity and reliability of the results. This section discusses those factors and how the project results were affected.

7.3.5 Reliability related to literature

To ensure the reliability of the literature, peer-reviewed articles published in established academic journals or books were selected for the literature review and desktop research. More recent publications were prioritised to ensure that the information was up-to-date and relevant to the context. In addition, official websites of relevant companies were used for descriptive purposes, for example to describe their activities. No claims or statements from these sources were used as evidence for arguments, to avoid potential bias or lack of objectivity.

7.3.1 Reliability related to interviews

The reliability in terms of the interviews was not very high as the interview guides were not published in this report, which means that it is not possible to recreate the exact same interviews in further work on this project. The reason they were not published was that no generalised interview guides were created, but instead all were tailored to each individual respondent as they were interviewed for different purposes. All respondents came from different backgrounds, worked in different industries, and therefore possessed different types of information and knowledge, which was the goal as different types of information were sought for each interview. Since a total of 29 interviews were conducted, no interview guides were published as it is a very high number, and it would be excessive information. However, it is probably not particularly necessary to carry out exactly the same types of interviews in further work as the main purpose of them was groundwork for the project, which means that they probably do not need to be recreated to take the project further. The validation interviews will in any case need to be remade when new iterations of the concepts are created, and new interview guides with questions adapted to them will be needed. This was the reason for excluding the interview guides, which also made it possible to keep the report shorter and exclude redundant information for the reader.

7.3.2 Validity related to interviews

Validity in relation to the interviews was relatively high. The fact that specific interview guides were written for each interview, instead of a standardised version, enabled the interviews to be more qualitative than quantitative, which was the overall aim. What weakened the validity was that some interview guides, mainly from the research phase, were in retrospect not optimal for obtaining the desired information. One mistake made was that analyses of what the potential answers to the questions could contribute were not made. The questions were only formulated according to what could be interesting and related to the project. In other words, the formulation was created on the basis of what was wanted to be asked about, not on the basis of what types of answers were wanted. As a result, several interviews did not lead to as much information as

desired. Many respondents gave very similar answers, which became redundant. This was an analysis made during the innovation phase, and therefore this was taken into account and the interviews became much more valuable after that.

7.3.3 Validity related to assumptions

Several assumptions, as described in Section 5.4.5, were made during the project due to various reasons. The main reason was the combination of the large number of activities that needed to be carried out, and the time limit to which the project scope needed to be adapted. Priorities needed to be made to ensure that the most crucial data could be collected. Another reason for why assumptions were made was due to that some respondents were either not willing to share information, especially related to their finances, or they did not have the necessary knowledge to provide answers. Also, some individuals which would have been necessary to interview to make more accurate estimations were not able or interested in participating in an interview.

These factors resulted in assumptions and rough estimates by the project members of varying degrees of certainty, depending on the amount of collected data. The least qualified assumptions were made due to almost a complete lack of data and almost made in its entirety on logical thinking. This caused an uncertainty into the financial calculations, which affected the validity of the results. However, since the project ended at an early stage of the future possible development, there was no alarming harm in the figures not being 100% accurate at this stage. In addition, the industrial supervisor emphasised early on in the project that rough estimates would be sufficient to provide a satisfactory picture of the feasibility and profitability of the concepts.

7.3.4 Financial validity

An initial objective was to calculate the ROI, mainly for the reference company but also for other actors. However, that objective could not be met, since the only calculation that could be made was how much the reference company's customers would have to pay in order for the reference company to earn a return after three years, i.e. a proper calculation of ROI was not performed. A better approach that could have led to actual ROI calculations would have been to present price proposals to potential customers during interviews, so that they could confirm whether they were willing to pay or not. This in turn would have led to estimates of profit and thus ROI calculations. Unfortunately, due to the fact that all the data needed to provide price proposals could not be obtained for the same reasons as why assumptions needed to be made as described in Section 7.4.3, a calculation of the ROI could not be made in this way. The only calculation that could be performed was therefore based on an estimated number of years that could be reasonably expected to realise a return on the sum of some of the reference company's larger costs. This affected validity as much of the necessary information was missing, which therefore means that the estimate of what the price for the reference company customers would be was very rough. Nevertheless, the figures that could be produced provide some indicators that may be useful in further work on the project.

7.4 Recommendations for future project development

This project will have to be further developed in order to reach a fully satisfying result where more aspects are considered in this complex innovation ecosystem.

7.4.1 The next steps in the concept development

Further work on the project can take many different directions, depending on priorities, resources and other external factors. Based on the project results, some recommendations have been made for certain steps that should be given high priority.

First of all, more logistics companies need to be interviewed. As they are an actor affected in all business models and also the customer in two of them, it is very important to get their opinions on the concepts, their roles, the transition to off-peak deliveries and whether they are willing to pay for the service. The fact that logistics companies are interested in co-loading and overnight deliveries does not mean that they would be willing to pay for it. Therefore, in order to achieve adoption, it is also important to gather financial data from logistics companies in order to calculate their profitability.

As goods recipients were the stakeholder identified as the most difficult to create adoption with, many actions are required to address this. To start with, more interviews need to be conducted with several shops, restaurants and cafés to get more validation of their problems and needs in order to then address their concerns about the concepts As system changes can be abstract and sometimes difficult to understand how to actually be affected in practice, the recommendation of this project is that many small-scale pilot projects should be carried out where stakeholders can more easily understand how they would be affected. Successful pilots will in turn lead to proof of concept, facilitating the scaling up of the business.

Once more research has been done and more data has been collected, new more accurate financial calculations need to be made. The calculations made in this project were only estimates of the main costs for the reference company and for the customer buying the service. However, as this service is a systemic innovation where money flows between several actors, these figures are a high priority to be produced in order to ensure adoption.

Some actors that have been identified as relevant to consider, both for the adoption aspect and for the economic viability of the concept, but which have been outside the scope of this project are: *IT solution providers* – for the storages and to enable transparency and communication between the parties sharing the data, *insurance companies* – to increase the understanding of how to insure the storage and how much it would cost, and *unions* representing the goods recipients – to evaluate their views on the concepts and how they would affect the goods recipients.

7.4.2 Possible future new value propositions

Further concept development that could bring new types of values were discovered but not further investigated in detail. These would be valuable to incorporate in the calculations and further formulation of the concepts.

Integrating solar panels for renewable electricity to the storages, as well as selling excessive electricity, which could be sold as a new revenue stream.

Instead of trying to hide the storage units if they are placed on municipal land if they are considered ugly, the opposite could be done by placing them even more visibly and promoting the sustainable development in logistics that these storages would bring, where the actors involved can promote themselves as being part of this initiative.

Improve the design of the storages while fostering creativity by implementing goodwill projects with, for example, schools, who can create artwork on them. This in turn leads to a more beautiful environment.

Lastly, another identified future possibility would be data collection of the influx and outflux of goods with purposes of optimising the storage solutions for the customers and creating better and better storage solutions over time.

7.5 Recommendations for future research on system innovation

After the finalised literature review on system innovation, it was concluded that there is a lot of literature on topics related to the concept and that many articles in fact are about system innovation, but the term was not used. It was possible to understand from the context that it was systems innovation that much of the literature was about, but it required knowledge of the concept to make that connection. Therefore, it is recommended that the term *system innovation* should be used more in the literature when writing about the topic to make it more widely recognised.

8. Conclusions

This report provides several useful academic research contributions and practical contributions by reaching both the academic and practical purposes, which provided answers to all three formulated research questions. Adoption of system innovations by involved stakeholders were found to be facilitated by several important factors, but it is different how easy it would be among different stakeholders, since they have different needs, values, problems, and conditions. The specific case of intermediate storage of refrigerated goods enabling off-peak deliveries in urban environments was found to probably be a desirable, feasible, viable, and sustainable business if an orchestrator offers a modular product/service-hybrid system due to the complexity of the varying factors in the system. Furthermore, some aspects regarding restrictions, necessary future developments, and other conditions need to first be addressed for this to be achieved completely. Difficulties were experienced in accessing data from involved actors, which made it difficult to evaluate the potential economic profitability of the concepts at a system level and for each actor within the system.

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Appendices

Appendix A: First iteration of concepts – sketches

Appendix B: Second iteration of concepts – sketches

Appendix C: Elaborate problem analysis

Appendix D: Elaborate concept solution hypotheses

Appendix E: Identified transactions in the concepts' new logistics chain

Appendix F: Problem and concept evaluation grading

Appendix G: Prompts in ChatGPT to produce AI-visualisations

Appendix A: First iteration of concepts – sketches

In this appendix, the first iteration of concepts' sketches can be viewed.

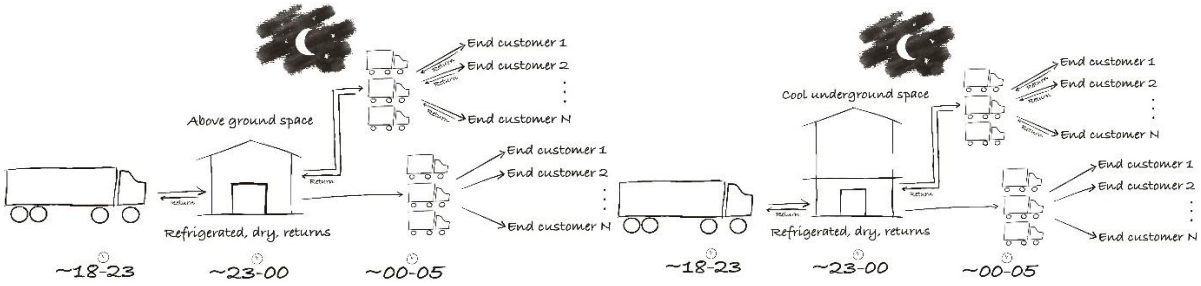


Figure A 1. Initial concept 1 – Utilisation of existing facilities (Image to the left: Today, Image to the right: Future.

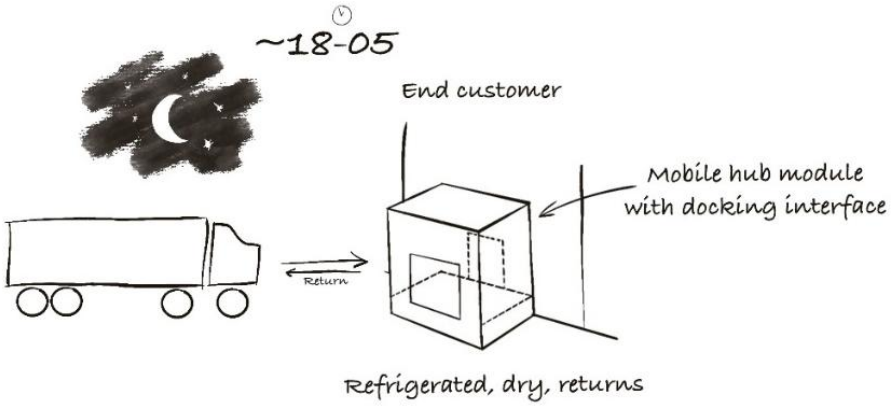


Figure A 2. Initial concept 2 - Mobile microhub gateway "slussen".

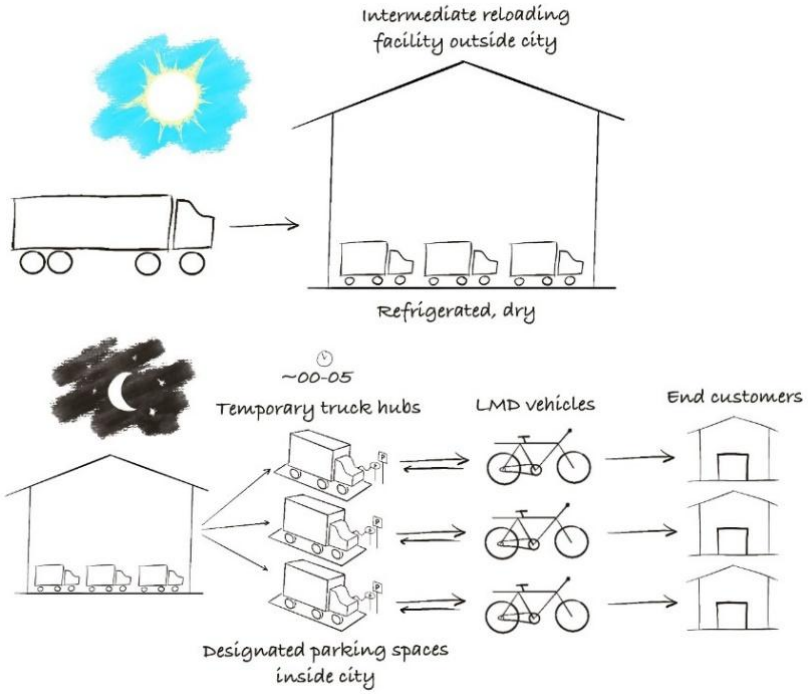


Figure A 3. Initial concept 3 - Autonomous trucks as a consolidation hub

Appendix B: Second iteration of concepts – sketches

In this appendix, the second iteration of concepts' sketches can be viewed.

Concept A1

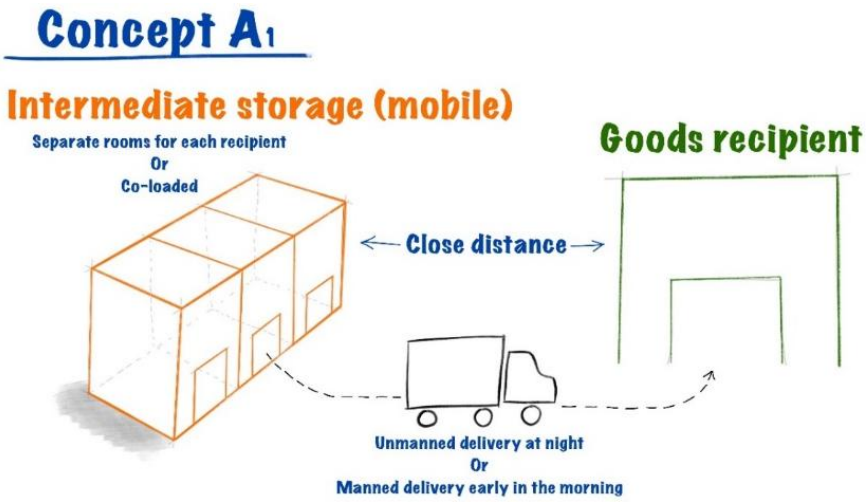


Figure B 1. Sketch of Concept A1 – Intermediate storage, LMD, and goods recipient.

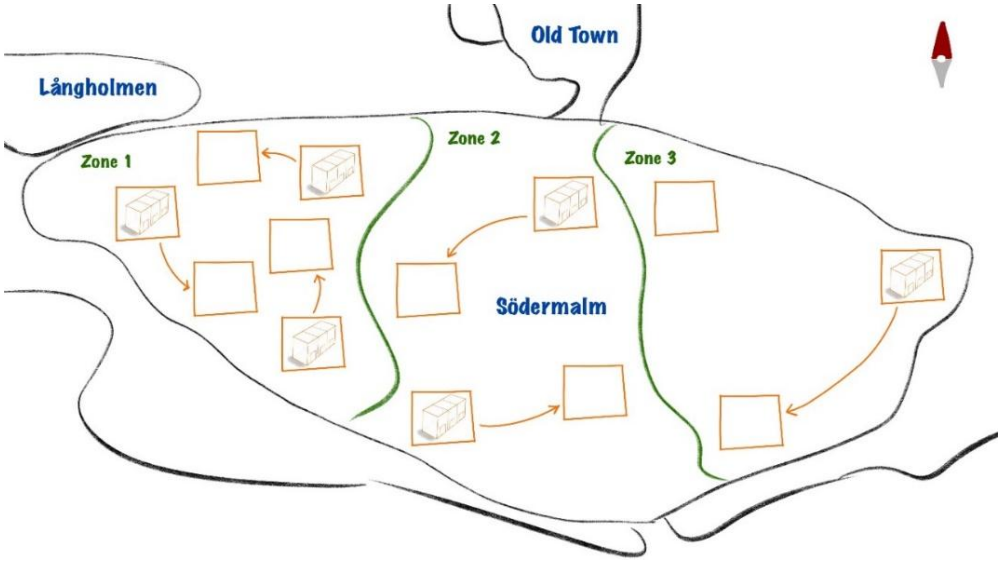


Figure B 2. Sketch of Concept A1 – Zones and hypothetical intermediate storage placement areas.

Concept A2

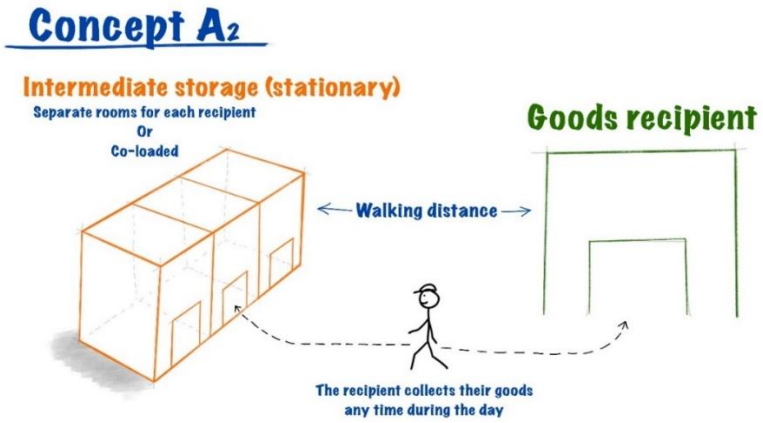


Figure B 3. Sketch of Concept A2 – Intermediate storage, LMD, and goods recipient.

Concept B1

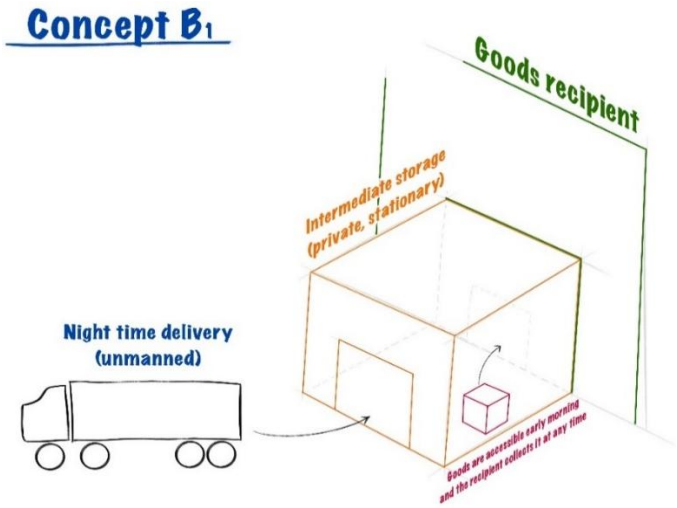


Figure B 4. Sketch of Concept B1 – Transporter, intermediate storage, and goods recipient.

Concept B2

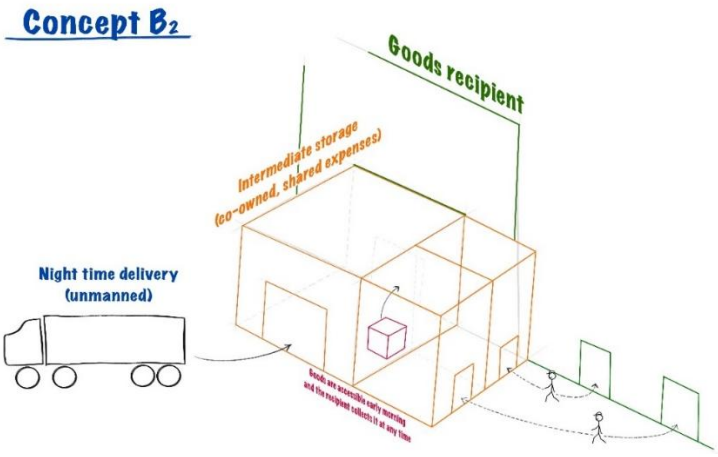


Figure B 5. Sketch of Concept B2 – Transporter, intermediate storage, primary goods recipient, and secondary goods recipients.

Appendix C: Elaborate problem analysis

In this appendix, a more elaborate problem analysis can be viewed.

Table C 1. Problem analysis and affected actors

Problem	Problem owner	Directly affected	Why does the problem exist?
Large number of orders = High load in the supplier's consolidation warehouse	Logistics company	Logistics company Goods supplier	* Higher demand than capacity * Miscalculation of seasonal goods * Business expansion leads to physical warehouse capacity constraints * Problems with staffing * Unable to take on new customers
Large individual orders requiring a separate delivery route cause sub-optimal routes, if instead more orders could have been loaded together in the same truck, which may require more vehicles or longer distances	Logistics company	Logistics company	* Several trucks will travel the same route, which is costly * Lack of more vehicles -> requires longer distances * Vehicle restrictions can lead to uneven fill rates * Geographically sub-optimal location of warehouses * Most often happens when the customer requires super specific times for deliveries = contractual)
Access restrictions while driving	Logistics company	Logistics company Driver	* External delays that cannot be influenced (snowstorm, demonstrations) * Traffic congestion * Deviation from planning, may cause suboptimal timing of delivery
Lack of loading space directly adjacent to the building where the delivery is made through an entrance crossing a public surface (pavement, cycle path, etc.)	Logistics company Real estate company	Logistics company Real estate company Driver Citizens	* Properties without an associated loading bay * Busy car park / loading zone * Difficult unloading
Damaged unsaleable goods / Broken cold chain	Logistics company	Logistics company Driver Grocery store Restaurant	* Incorrect packaging of goods * Incorrect sorting of goods

		Café	* Recipient does not receive saleable goods * Cold chain has not been maintained
Repeated sufficiently late deliveries by the recipient (on an annual basis)	Grocery store Restaurant Café	Logistics company Driver Grocery store Restaurant Café Citizens	* Goods recipients do not receive their deliveries on time * Causes problems in logistics companies' planning * Chain reaction with delayed deliveries
Manual reception of goods (Boom drive due to recipient not on site, assistance during reception)	Grocery store Restaurant Café	Logistics company Driver Grocery store Restaurant Café	* Recipients not present at delivery * If deliveries require manual reception, these cannot take place off-peak * Recipients need to assist with delivery
Incorrect delivery	Logistics company	Logistics company Driver Grocery store Restaurant Café Wrong recipient	* Recipients do not receive their goods on time * Costly handling
Unsustainable strain on staff	Grocery store Restaurant Café	Logistics company Grocery store Restaurant Café Consumer	* Delayed deliveries cause scheduling problems * Too few staff need to perform too many tasks * Extra costs for goods recipients
Costly storage space	Grocery store Restaurant Café	Grocery store Restaurant Café	* Unutilised warehouse space = waste of money * Takes up shop space
Expensive capital-intensive inventory	Grocery store Restaurant Café	Goods supplier Grocery store Restaurant Café	* Consumption not equal to estimated * Occupies warehouse space = costly (stocking and alternative sales)
Not having goods to sell	Grocery store Restaurant Café	Goods supplier Grocery store Restaurant Café Consumer	* Goods have not arrived with delivery * Lost revenue * Negative customer experience
Not having goods to buy	Consumer	Grocery store Restaurant Café Consumer	* Negative customer experience * Grocery stores do not have goods on the shelf * Restaurants/cafés do not have dishes on the menu

Appendix D: Elaborate concept solution hypotheses

In this appendix a more elaborate concept solution hypotheses can be viewed.

Table D 1. Solution table for concept A, when off-peak deliveries and storage has been implemented

Problem	Solution hypothesis	Potential consequential problems	Specific measures in the solution	Potential economic beneficiaries	Why do they benefit economically
Large number of orders = High load in the supplier's consolidation warehouse	Unburdening the consolidation centre by allowing goods to be transported directly to the intermediate storage facility	Potentially new costs connected to more storage space	Utilisation of intermediate storage as additional storage space for supplier	Logistics companies	Benefit from being able to expand operations and their main warehouse space faster.
Large individual orders requiring a separate delivery route cause sub-optimal routes, if instead more orders could have been loaded together in the same truck, which may require more vehicles or longer distances	The concept shortens distances and reduces the number of vehicles	More outflows from the consolidation centre	The location of the warehouse is closer to the recipient	Logistics companies	Lower transport costs and personnel costs
Access restrictions while driving	Larger time margins due to off-peak deliveries		Placement of storage and enabling deliveries to be made when there are less access restrictions	Logistics companies	Reduced risk of financial compensation to recipients due to delayed deliveries
Lack of loading space directly adjacent to the building where the delivery is made through an entrance crossing a	Gives better opportunity for the driver to fulfil his mission from suppliers		Off-peak deliveries make existing lack of optimal loading space less of a problem	Logistics companies	More efficient offloading saves time and therefore money

public surface (pavement, cycle path, etc.)	Potentially not perform LMD = Problem disappears for supplier				
Damaged unsaleable goods / Broken cold chain	Detect the problem faster due to control of goods in intermediate storage Off-peak gives more time to fix the problem	Risk of more damaged goods due to more intermediaries and transhipments New costs linked to retrieving and replacing the damaged goods	Possibility to check the goods during transhipment at intermediate storage	Logistics companies	Potentially no financial compensation to recipients due to damaged goods
Repeated sufficiently late deliveries by the recipient (on an annual basis)	More efficient off-peak deliveries reduce the risk of delays. If errors occur during delivery, off-peak gives more time margin to fix the error	Risk of errors related to transhipment/incorrect delivery can lead to even more inefficient deliveries	Less risk of delays due to more efficient and earlier deliveries for those customers who can receive the goods during off-peak	Logistics companies Grocery stores Cafés Restaurants	Grocery stores, restaurants, and cafés does not lose income on lost sales. Logistics companies potentially do not have to provide financial compensation to recipients due to delayed delivery
Manual reception of goods (Boom drive due to recipient not on site, assistance during reception)	No risk of boom runs if goods can be received unmanned. Concept A2 enables them not to have to adapt to suppliers = manual reception becomes a non-issue	Boom driving may still occur if manual reception still has to take place Potential problems in concept A2 at own pick-up	Location of intermediate storage in concept A2 is close enough to the recipient to allow them to collect the goods themselves. Unmanned reception in concept A1 eliminates the problem	Logistics companies Grocery store Restaurant Café	Concept A1 eliminates risk of extra costs linked to boom runs. Shop / Restaurant in concept A2 gets lower delivery cost as they perform LMD themselves

Incorrect delivery	Potentially fewer wrong deliveries due to extra tracking system Greater time margin to fix wrong delivery	Risk of more wrong deliveries due to more intermediaries and transhipments If wrong deliveries are not detected by the driver, they will be detected later than today, making actions taken later and more costly in concept A2	Potentially shorter distances to retrieve new goods in case of damage if buffer is available in the storage Potentially shorter mileage to correct mis-driving	Logistics companies Grocery store Restaurant Café	Grocery stores, restaurants, and cafés get the right delivery faster. (Less risk of drivers being financially penalised)
Unsustainable strain on staff	More punctual deliveries mean less strain on staff time Possibility of self-collection from storages	Risk of physical strain	Easier to schedule staff and reduced risk of overtime Concept variant A1 provides less risk of physical strain (due to correct punctual delivery)	Grocery store Restaurant Café	Grocery store, restaurant, and café can potentially have lower labour costs due to less overtime.
Costly storage space	Possibility of peak season buffer stock to compensate for the cost of the extra storage space	The cost of the extra stock and storage space	If the grocery store, restaurant, or café does not currently have the possibility to order and stock seasonal goods, they have a greater opportunity to do so with concept variant A2 in particular, depending on the situation	Grocery store Restaurant Café	More opportunity to sell more seasonal goods as they can have more in stock (increased revenue)
Expensive capital-intensive inventory	Possibility of peak season buffer stock to compensate for the cost of the extra storage space	The cost of the extra stock and storage space	If the grocery store, restaurant, or café does not currently have the possibility to order and stock seasonal goods, they have a greater opportunity to do so with concept variant A2 in particular, depending on the situation	Grocery store Restaurant Café	More opportunity to sell more seasonal goods as they can have more in stock (increased revenue)

Not having goods to sell	Less risk of goods not being available due to more reliable deliveries. Less dependence on supplier in concept A2 as they collect the goods themselves	May be more risk of errors linked to intermediaries and transshipment	No empty shelves due to more reliable early delivery Possibility to pick up the desired volume and restock shelves during the day with concept A2	Grocery store Restaurant Café	Grocery store, restaurant, and café gets retained clientele, satisfied customers and no lost sales and the possibility of increased sales since possibility to replenish during the day
Not having goods to buy	Less risk of goods not being available due to more reliable deliveries	May be more risk of errors linked to intermediaries and transshipment	Not having empty shelves due to more reliable early delivery	Grocery store Restaurant Café	Grocery store, restaurant, and café retains clientele, satisfied customers and no lost sales

Table D 2. Solution table for concept B, when off-peak deliveries and storage has been implemented

Problem	Solution hypothesis	Potential consequential problems	Specific measures in the solution	Potential economic beneficiaries	Why do they benefit economically
Large number of orders = High load in the supplier's consolidation warehouse	Unchanged				
Large individual orders requiring a separate delivery route cause sub-optimal routes, if instead more orders could have been loaded together in the same truck, which may require more vehicles or longer distances	Unchanged				
Access restrictions while driving	Larger time margins due to off-peak deliveries		Enabling deliveries to be made when there are minor access restrictions	Logistics companies	Reduced risk of financial compensation to recipients due to delayed deliveries
Lack of loading space directly adjacent to the building where the delivery is made through an entrance crossing a public surface (pavement, cycle path, etc.)	Provides better opportunity for the driver to carry out his assignment from suppliers due to standardised unloading		Standardised interface	Logistics companies	More efficient offloading saves time and therefore money
Damaged unsaleable goods / Broken cold chain	Checking goods at night reduces the risk of delivery of damaged goods	Unmanned reception increases risk of delayed detection of damaged	Some type of technical solution allowing for the control of	Logistics companies Grocery stores Restaurants Cafés	Potentially no financial compensation to recipients due to

	and the time margin to fix the error	goods if driver makes errors	goods and accountability		damaged goods If driver detects damaged goods and manages to fix it in time, shop / restaurant benefits as they do not have unsaleable goods
Repeated sufficiently late deliveries by the recipient (on an annual basis)	Off-peak deliveries reduce the risk of delays		Ensures that goods arrive on time because the hub is located directly next to the recipient	Grocery stores Restaurants Cafés	They have a better chance of always having goods to sell, which reduces the risk of lost sales
Manual reception of goods (Boom drive due to recipient not on site, assistance during reception)	No risk of boom driving when goods are received unmanned Unmanned reception means that recipients do not need to assist with delivery	If there is a technical problem with the storage when it is unmanned, delivery will not be possible	Digital two-factor authentication provides assurance to recipients that goods have been delivered - manual receipt becomes a non-issue	Logistics companies Grocery stores Restaurants Cafés	Logistics companies save time and resources by eliminating the risk of extra costs associated with boom runs Grocery stores, restaurants, and cafés do not need to have staff on site for reception
Incorrect delivery	Greater time margin to correct incorrect delivery	If incorrect deliveries are not detected by the driver, they are detected later than today, which means that actions are taken later and are more costly	Digital two-factor authentication to ensure the right goods have been delivered to the right storage	Grocery stores Restaurants Cafés	Reduced risk of handling costs in case of incorrect deliveries (lost sales for Grocery stores, restaurants and cafés

Unsustainable strain on staff	More punctual deliveries mean less strain on staff time		Ensures that goods arrive on time because the hub is located directly next to the recipient	Grocery stores Restaurants Cafés	They do not need to have extra staff assigned to reception
Costly storage space	Possibility of peak season buffer stock to compensate for the cost of the extra storage space	The cost of the extra stock and storage space	Possibility to expand the storage area when larger stocks are needed (Peak seasons)	Grocery stores Restaurants Cafés	Opportunity for increased sales that will hopefully offset the cost
Expensive capital-intensive inventory	Possibility of peak season buffer stock to compensate for the cost of the extra storage space	The cost of the extra stock and storage space	Possibility to expand the storage area when larger stocks are needed (Peak seasons)	Grocery stores Restaurants Cafés	Opportunity for increased sales that will hopefully offset the cost
Not having goods to sell	Less risk of goods not being available due to more reliable deliveries Because goods are more readily available, they can restock shelves when needed		Ensures that goods are on the shelves on time because goods arrive in the storage on time	Grocery stores Restaurants Cafés	They benefit from maximum fill rate / stock holding which eliminates the risk of lost revenue
Not having goods to buy	Less risk of goods not being available due to more reliable deliveries		Ensures that goods are on the shelves on time because goods arrive in the storage on time	Grocery stores Restaurants Cafés	They benefit from maximum fill rate / stock holding which eliminates the risk of lost revenue

Appendix E: Identified transactions in the concepts' new logistics chains

In this appendix a visualisation of identified transactions along the concepts' new logistic chains can be viewed as well as which actors have the costs and what to calculate.

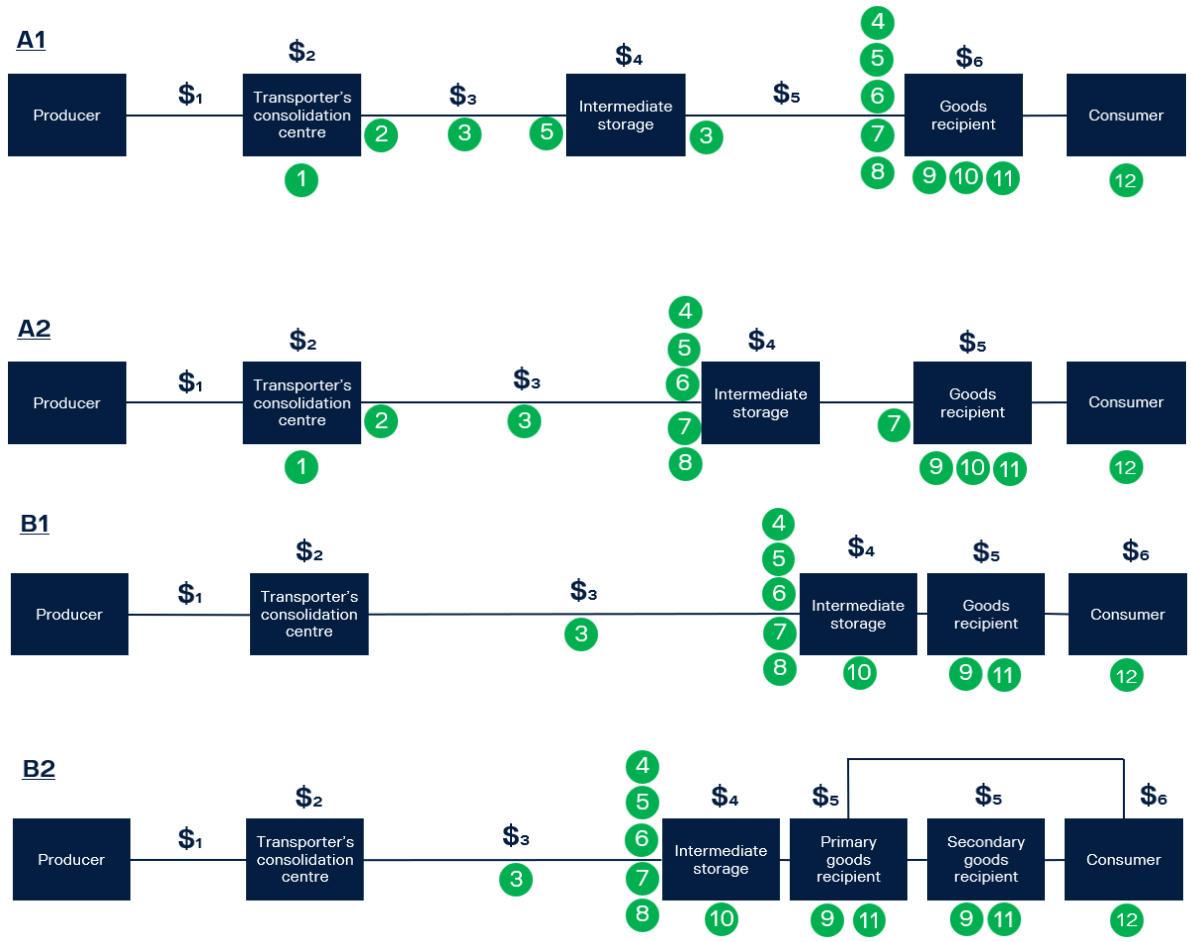
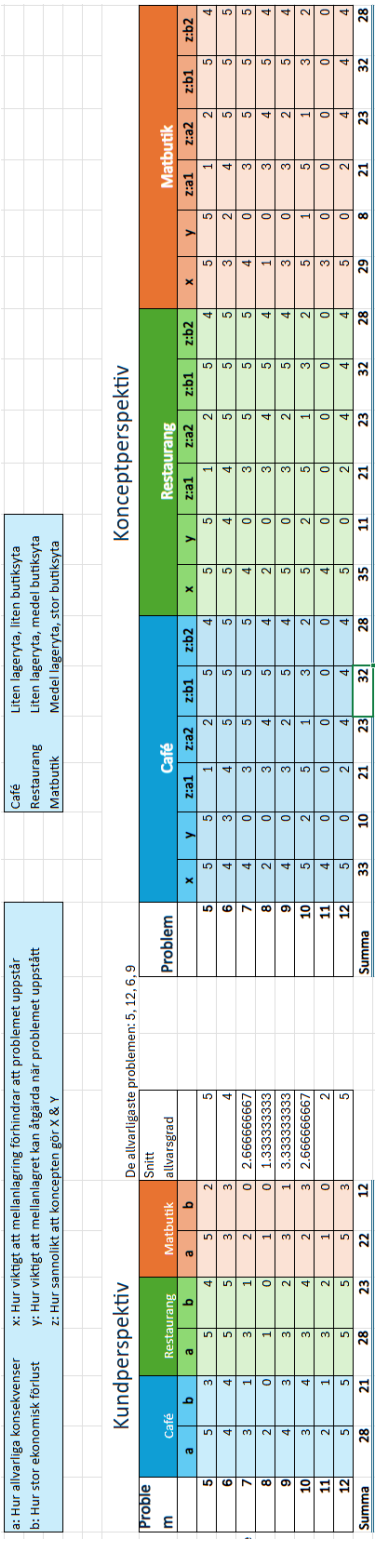


Figure E 1. Transaction mapping and solved problems of all concept variants.

Appendix F: Problem and concept evaluation grading



a: Hur allvarliga konsekvenser
b: Hur stor ekonomisk förlust

x: Hur viktigt att mellanlagring förhindrar att problemet uppstår
y: Hur viktigt att mellanlagret kan åtgärda när problemet uppstått
z: Hur sannolikt att konceptten gör X & Y

Café — Liten lageryta, liten butiksyta
Restaurang — Liten lageryta, medel butiksyta
Matbutik — Medel lageryta, stor butiksyta

Kundperspektiv

De allvarligaste problemen: 5, 12, 6, 9

Problem	Café a	Café b	Restaurang a	Restaurang b	Matbutik a	Matbutik b	Snitt allvarsgrad
5	5	3	5	4	5	2	5
6	4	4	5	5	3	3	4
7	3	1	3	1	2	0	2.666666667
8	2	0	1	0	1	0	1.333333333
9	4	3	3	2	3	1	3.333333333
10	3	4	3	4	2	3	2.666666667
11	2	1	3	2	1	0	2
12	5	5	5	5	3	5	5
Summa	28	21	28	23	22	12	

Konceptperspektiv

Problem	Café x	Café y	Café z:a1	Café z:a2	Café z:b1	Café z:b2	Restaurang x	Restaurang y	Restaurang z:a1	Restaurang z:a2	Restaurang z:b1	Restaurang z:b2	Matbutik x	Matbutik y	Matbutik z:a1	Matbutik z:a2	Matbutik z:b1	Matbutik z:b2
5	5	5	5	1	2	5	4	5	5	1	2	5	4	5	5	1	2	5
6	4	3	4	5	5	5	5	5	4	4	5	5	5	3	2	4	5	5
7	4	0	3	5	5	5	4	0	3	5	5	5	4	0	3	5	5	5
8	2	0	3	4	5	4	2	0	3	4	5	4	1	0	3	4	5	4
9	4	0	3	2	5	4	5	0	3	2	5	4	3	0	3	2	5	4
10	5	2	5	1	3	2	5	2	5	1	3	2	5	1	5	1	3	2
11	4	0	0	0	0	0	4	0	0	0	0	0	3	0	0	0	0	0
12	5	0	2	4	4	4	5	0	2	4	4	4	5	0	2	4	4	4
Summa	33	10	21	23	32	28	35	11	21	23	32	28	29	8	21	23	32	28

Figure F 1. Table of the second problem analysis for the grocery stores, restaurants, and cafés market segment. Note that the table is written in Swedish.

Appendix G: Prompts in ChatGPT to produce AI-visualisations

These are all the prompts that were used in ChatGPT to create the AI visualisations. All prompts were written by the project members, and they are listed in chronological order for each figure, with the last one generating the selected image. The prompts were written in Swedish and have not been translated to keep perfect accuracy.

Figure 4

”Perfekt! Bra jobbat! Kan du göra en till bild med exakt samma utformning men att lagret nu är placerat utomhus bredvid en tunnelbanestation som är ovanför markplan? Den ska vara placerad med baksidan emot en vägg, där den inte är i vägen för allmänheten. Gör det tydligt att tåg finns i bakgrunden. Vyn ska vara snett ovanifrån en bit bort.”

Note that this prompt was written directly after Figure 7 was created, which means that the generation of Figure 4 was based on the prompt for that figure.

Figure 5

“Perfekt! Bra jobbat! Kan du göra en till bild med exakt samma lager fast nu är den placerad inne i city bredvid trottoaren på en väg. Vägen ska vara som en mindre tvärgata där det inte är mycket trafik eller så mycket människor som rör sig. Bredvid vägen finns det hus som är typiska för centrala städer. Lagret får inte blockera någon dörr eller några fönster på husen. Jag vill att vägen ska vara en återvändsgränd, dvs, att vägen tar slut. Gör så att det syns tydligt.”

”Utformningen av lagret ser bra ut, behåll det, men kan du göra vägen bredare? Hela bredden av vägen behöver inte synas i bild. Placera även lagret närmare trottoarkanten. Den ska stå precis bredvid. Ta även bort fönstren som är bakom lagret.”

”Nej det där var inte bra. Ställ tillbaka lagret där den stod som på bilden innan, men nära trottoarkanten. Det ska bara vara en rak väg.”

“Den ser bra ut, behåll bakgrunden exakt som den är, och utformningen på lagret, men flytta lagret närmare trottoarkanten till vänster.”

”Nu står lagret för nära husväggen, flytta ut det lite längre. Sen vill jag att du zoomar ut vyn så man ser lagret på ett lite längre avstånd.”

Note that the first prompt was written directly after Figure 4 was created, which means that the generation for Figure 5 was based on the prompt for that figure.

Figure 7

”Jag vill att du gör en realistisk visualisering av ett lager som ser ut som en container, ca 6m långt och 3m brett. På ena långsidan av lagret ska det finnas tre likadana inkörsportar för godsleveranser som ser ut som klassiska rullportar med tydliga handtag i nedre delen av dörren. Lagret ska se ut som en klassisk plåtcontainer, ljusgrå, korrugerad plåt. Lagret ska vara placerad i ett parkeringsgarage intill en vägg där den inte är i vägen för bilar.”

Figure 8

“Jag vill att du gör en realistisk visualisering av ett lager som ser ut som en container. Lagret ska vara placerad utomhus i direkt anslutning, vägg i vägg, till en byggnad på lagrets ena

långsida. Lagret ska ha tre likadana inkörsportar för godsleveranser på väggen som är mittemot sidan mot byggnaden, alltså den andra långsidan. Lagret ska se ut som en klassisk plåtcontainer, ljusgrå, korrugerad plåt. Inkörsportarna ska se ut som klassiska rullportar med tydliga handtag i nedre delen av dörren. Vyn ska vara snett ovanifrån en bit bort. Det ska vara dagsljus. Lagret och byggnaden ska vara placerat i ett inhägnat område, som en innergård. Gör det tydligt att området används för att lasta av gods in till den röda byggnaden. Området är inte tillgängligt för allmänheten."

"Det är en bra utformning av lagret. gör en ny bild med samma lager, men placera den längre bak så lagret står vägg i vägg mot byggnaden bakom, med baksidan mot byggnaden. Gör också mer hus i bakgrunden istället för staket för att markera att det är en innergård."

"Väldigt bra bakgrund, perfekt! Men flytta lagret längre bak så det står vägg i vägg med byggnaden bakom den."

Figure 10

"Jag vill att du gör en realistisk visualisering av ett lager som ser ut som en container. Lagret ska vara placerad utomhus i direkt anslutning, vägg i vägg, till en byggnad på lagrets baksida, alltså ena kortsidan. Lagret ska ha en inkörsport för godsleveranser på väggen som är mittemot sidan mot byggnaden, alltså den andra kortsidan. Lagret ska se ut som en klassisk plåtcontainer, ljusgrå, korrugerad plåt. Inkörsporten ska se ut som en klassisk rullport med tydligt handtag i nedre delen av dörren. Vyn ska vara snett ovanifrån en bit bort. Det ska vara dagsljus. Lagret och byggnaden ska vara placerat i ett inhägnat område, som en innergård i stadsmiljö. Gör det tydligt att området används för att lasta av gods in till byggnaden bakom lagret. Området är inte tillgängligt för allmänheten. Få det att se ut som att det finns en lagerdörr bakom lagret som går in till byggnaden."

"Det var ganska bra. Men gör lagret mindre, ca 4x2x3 m. Ta bort dörren på byggnaden bredvid lagret och lägg istället till en större port bakom lagret. Det räcker att man ser utkanterna av den dörren. Bakgrunden är bra annars, behåll den men zooma ut så man ser på ett längre avstånd. Gör även stängslet högre. Lägg till ett handtag i nedre delen av dörren till lagret."

"Flytta bak lagret så det står direkt framför dörren in till byggnaden. Gör även lagret lite högre, ca 0.5m högre. Behåll bakgrunden som den är. Vyn ska va på ett längre avstånd snett ovanifrån. Lägg till ett handtag i nedre delen av lagerdörren."

"Lagret ska stå vägg i vägg med byggnaden bakom, framför dörren in till byggnaden. Zooma ut så man ser mer av bakgrunden som ser ut som en stadsmiljö."

"Bättre! Behåll exakt samma bakgrund och utformning av lagret, men flytta lagret längre bak så den är placerad precis framför dörren in till byggnaden. Du kan ta bort den upphöjda kanten på marken så du får en plan yta för lagret att stå på framför byggnaden."

Figure 11

"Jag vill att du gör en realistisk visualisering av ett lager som ser ut som en liten container, ca 2x3x2,5 m stor. Lagret ska vara placerad utomhus i direkt anslutning, vägg i vägg, till en byggnad på lagrets ena kortsida. Lagret ska ha en inkörsport för godsleveranser på väggen som är mittemot sidan mot byggnaden, alltså den andra kortsidan. Lagret ska se ut som en klassisk

plåtcontainer, ljusgrå, korrugerad plåt. Inkörsporten ska se ut som en klassisk rullport med ett tydligt handtag i nedre delen av dörren. Vyn ska vara snett ovanifrån en bit bort, utzoomat. Det ska vara dagsljus. Lagret och byggnaden ska vara placerat i ett inhägnat område, som en innergård. Gör det tydligt att området används för att lasta av gods in till den röda byggnaden. Området är inte tillgängligt för allmänheten.”

”Lagret ser bra ut och placeringen är bra, men kan du zooma ut så man ser från ett längre avstånd? kan du också göra om bakgrunden så det syns tydligare att det är ett inhägnat område som används för att lasta av gods, typ som en lastkaj?”

TRITA – ITM – EX 2025:200
Stockholm, Sweden 2025

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